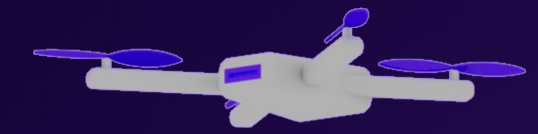
A Survivalist's Guide to .NET MAUI





David Ortinau

@davidortinau

davidortinau@noc.social

david.ortinau@microsoft.com

MODERN CLIENT

David Ortinau

Principal Product Manager

St. Louis, Missouri USA

Help customers succeed delivering client app experiences across various screens using .NET SDKs for mobile and desktop cross-platform – aka .NET MAUI (and Xamarin).

I like to trail run and have completed multiple distances up to 50 miles.

Grandfather. Doesn't age.

.NET ecosystem momentum

5.7M

Monthly active .NET developers

In Visual Studio family

#1

Most Loved Framework (2019, 2020, 2021)

.NET Core

Top 30

Highest velocity OSS project

github.com/dotnet

Top 5

Language on GitHub

C#

>10X

Faster than Node.js

ASP.NET Core

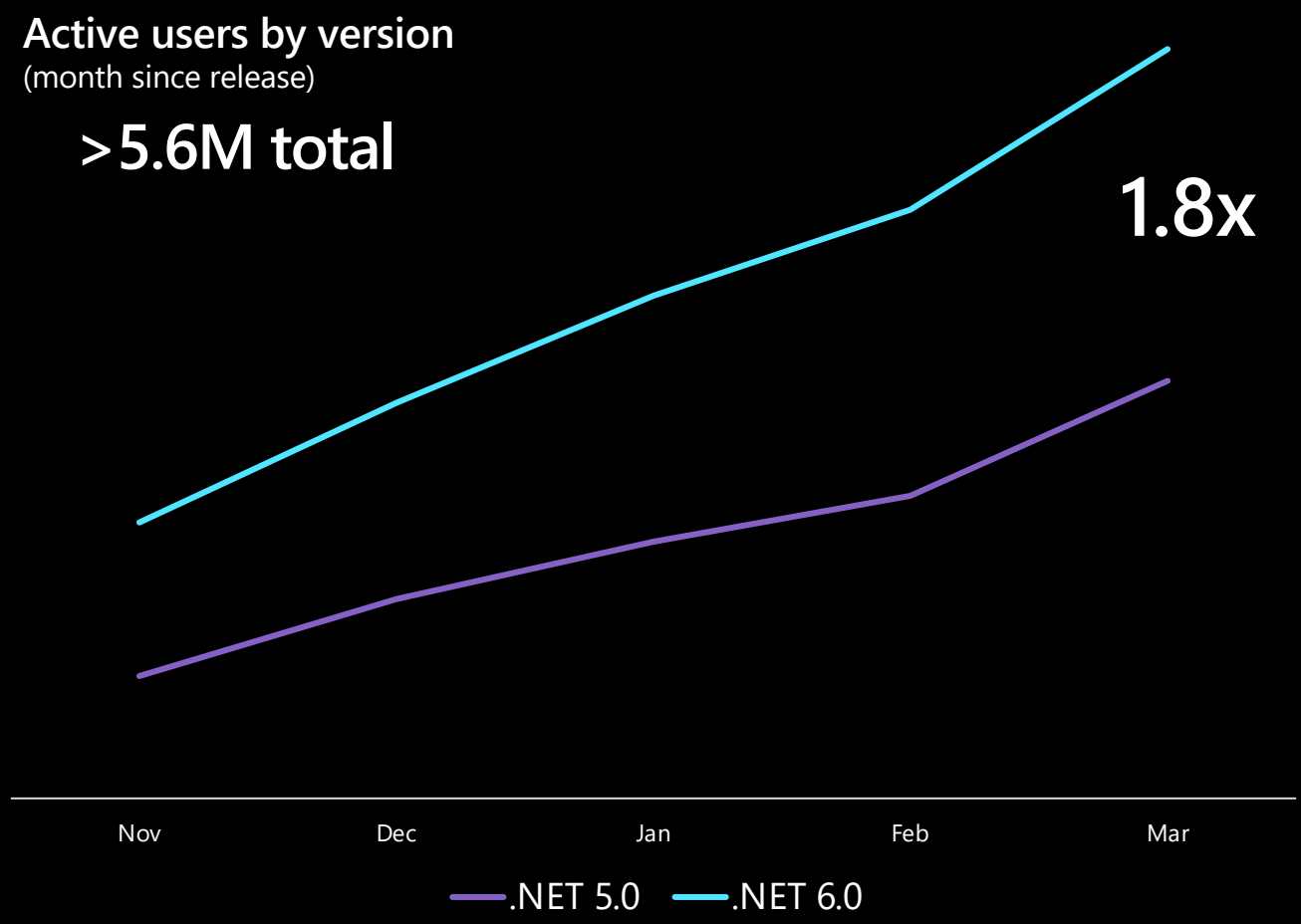
40%

New to .NET are students

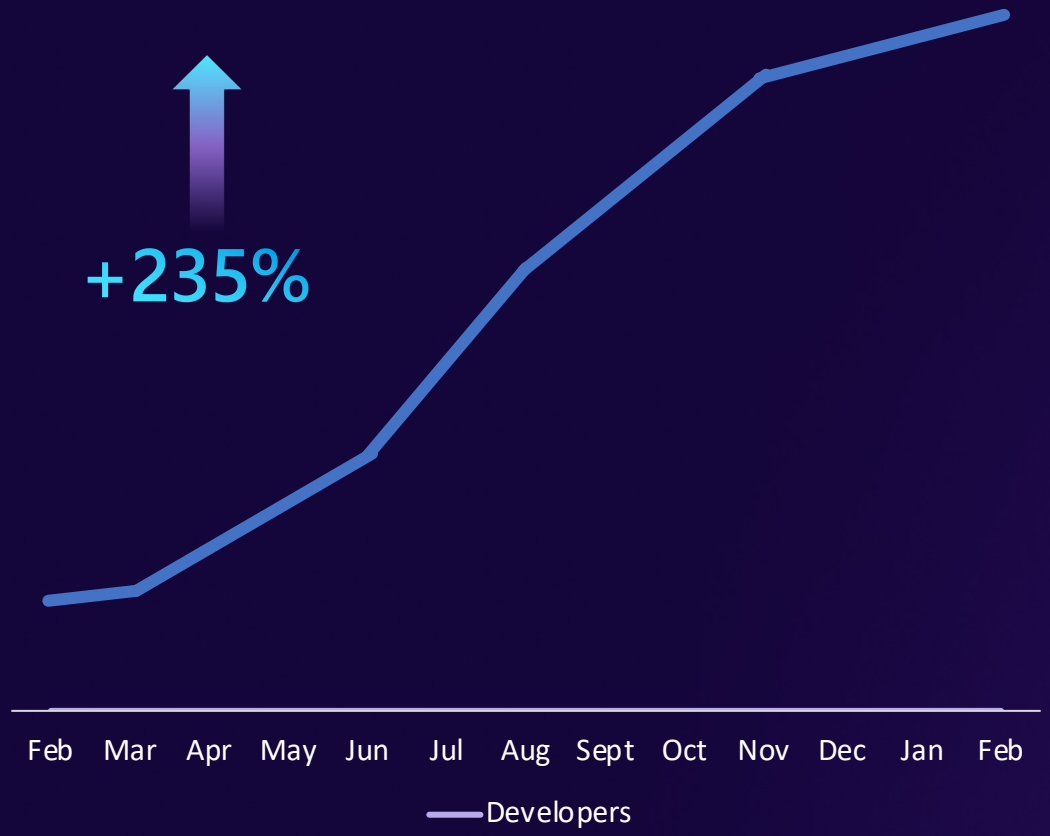
Download survey

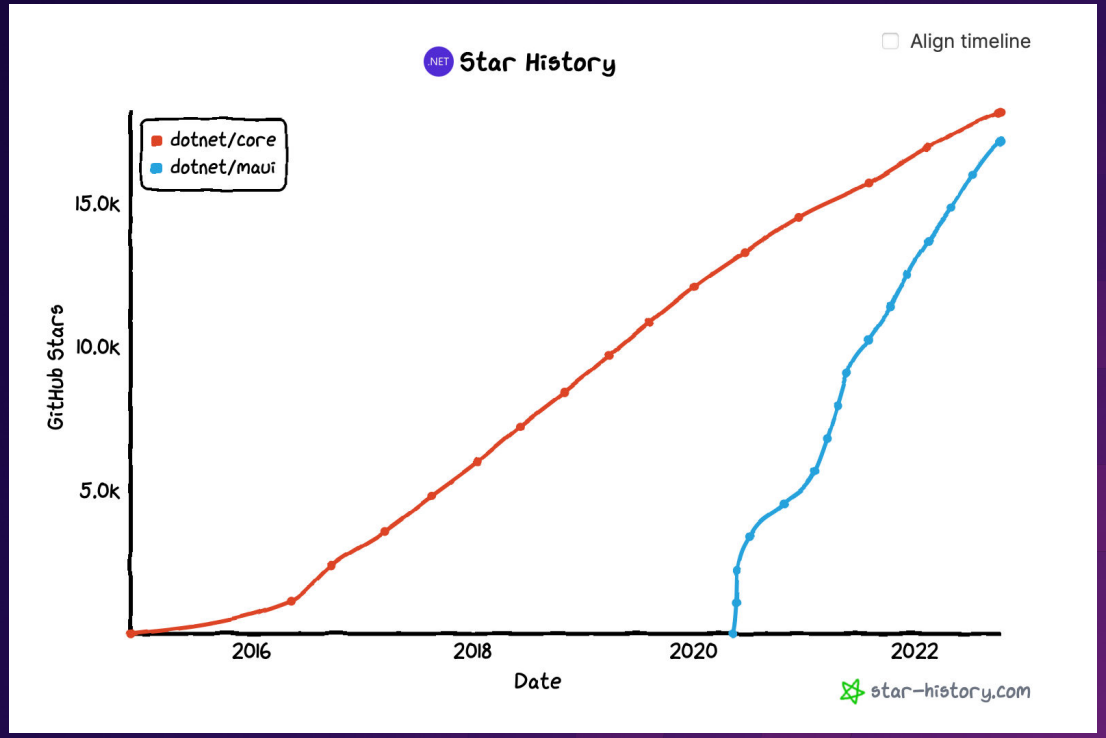
Sources: insights.stackoverflow.com/survey/2019#most-loved-dreaded-and-wanted; insights.stackoverflow.com/survey/2020#most-loved-dreaded-and-wanted; github.com/cncf/velocity/blob/master/reports/top_projects_by_activity.txt; octoverse.github.com/; www.techempower.com/benchmarks/#section=data-r20&hw=ph&test=plaintext

.NET 6 adoption



.NET MAUI Growth





M365 SharePoint Admin

.NET for Android and iOS

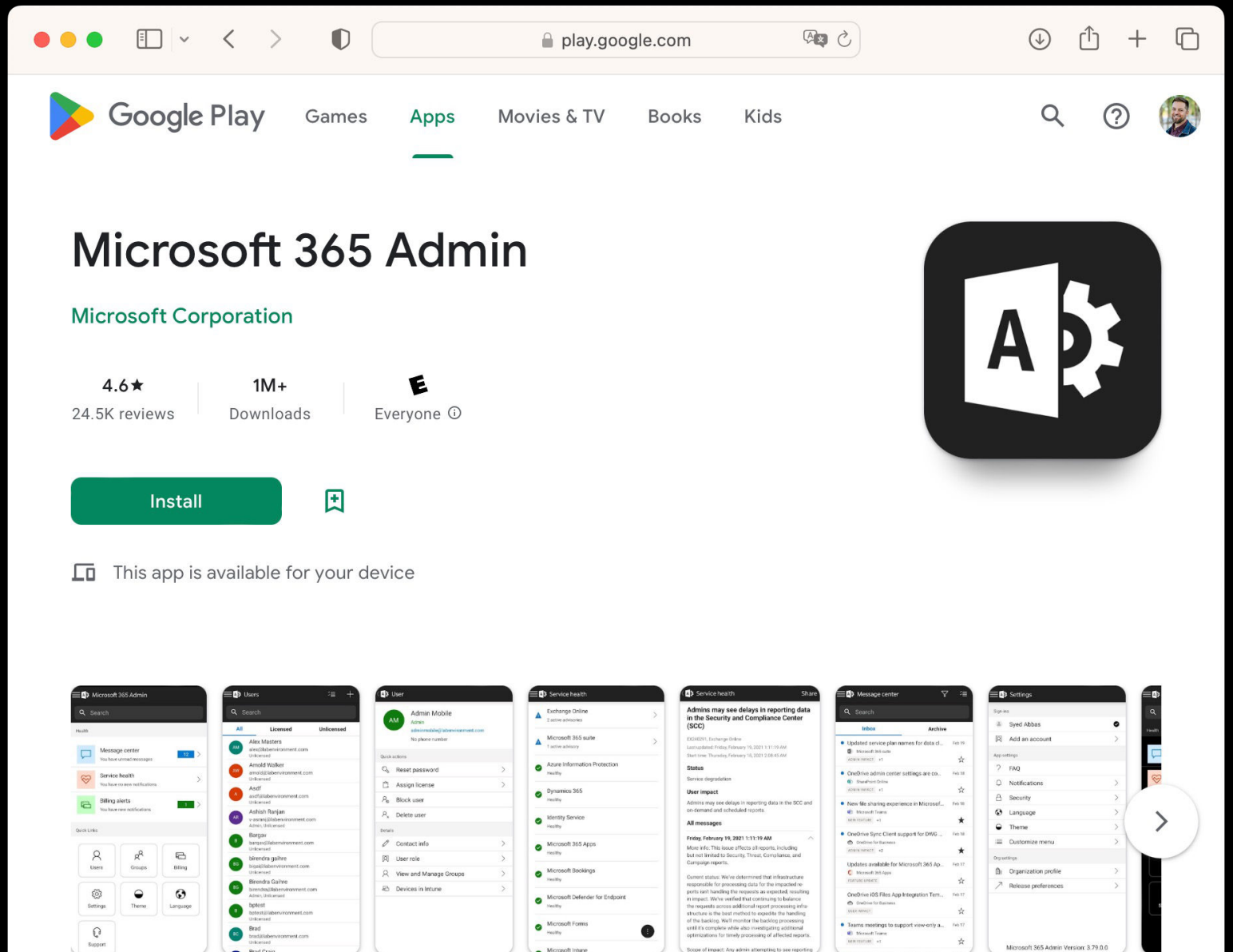
Azure

- Active Directory
- Notification Hub

App Center

- Analytics
- Crash reporting
- Distribution
- Test

InTune



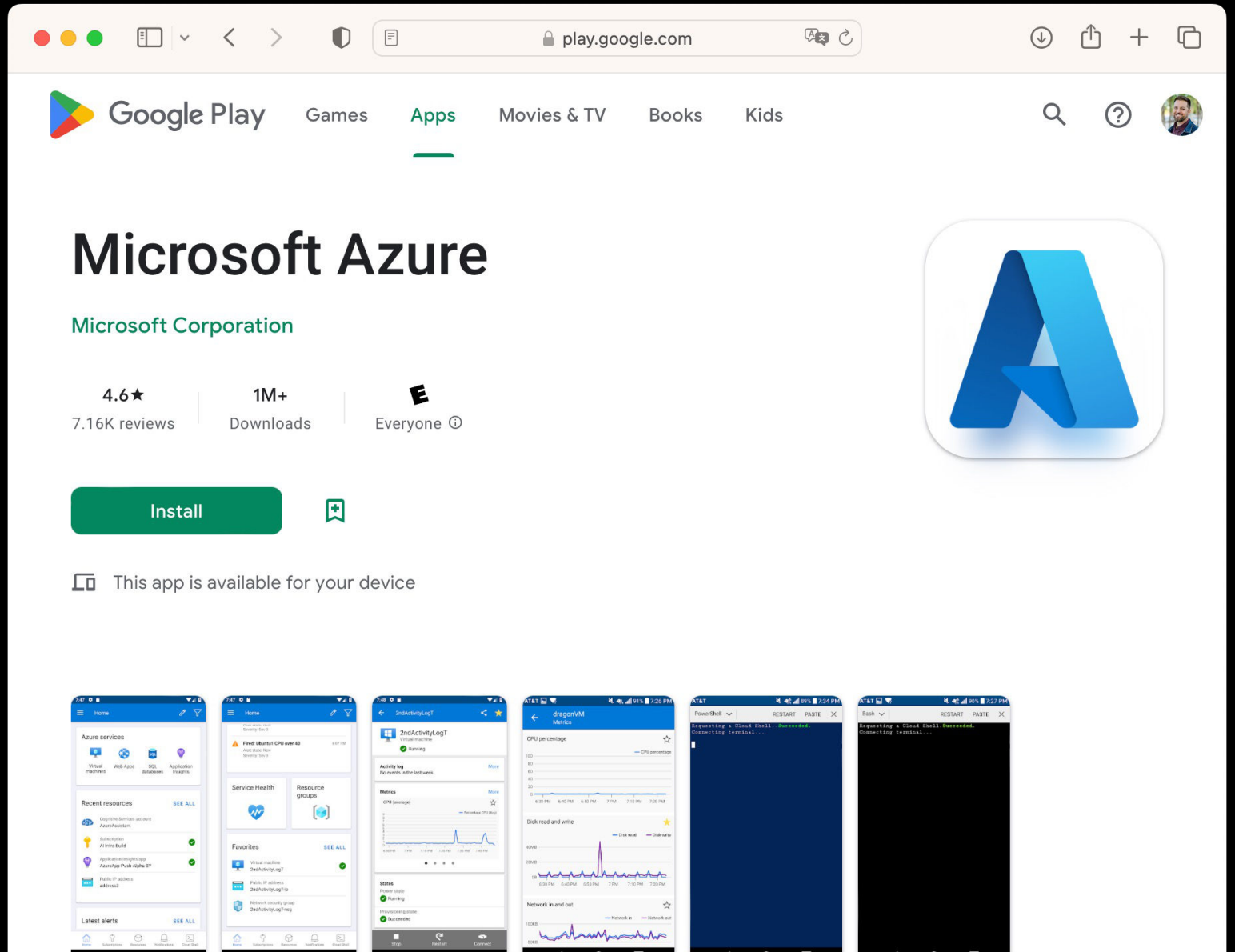
Azure Mobile

.NET for Android and iOS

Azure

- Active Directory
- Blob storage
- Notification Hub

Cosmos DB



SeeingAI

.NET

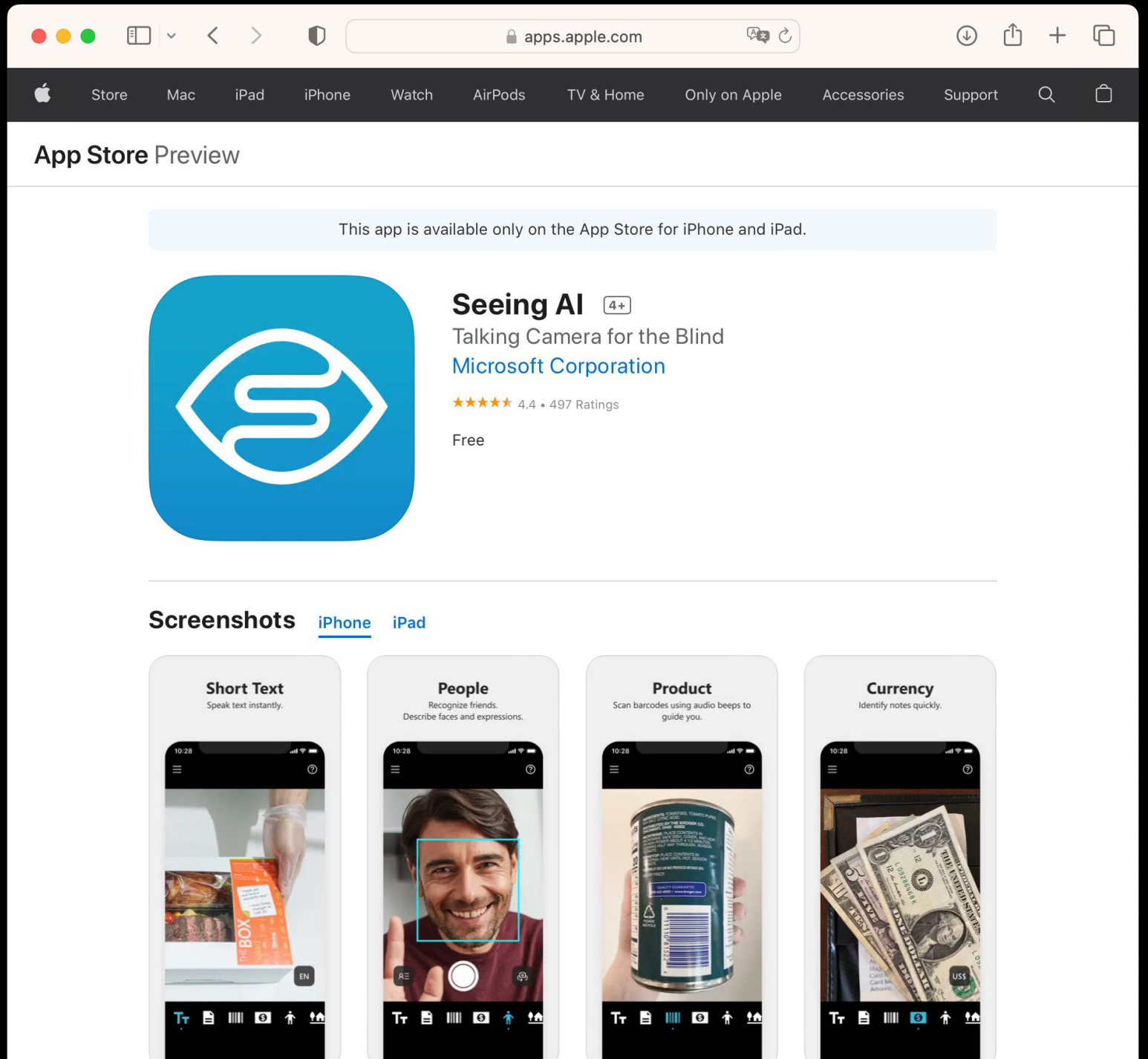
- iOS
- Android in development

App Center

- Analytics
- Crash reporting

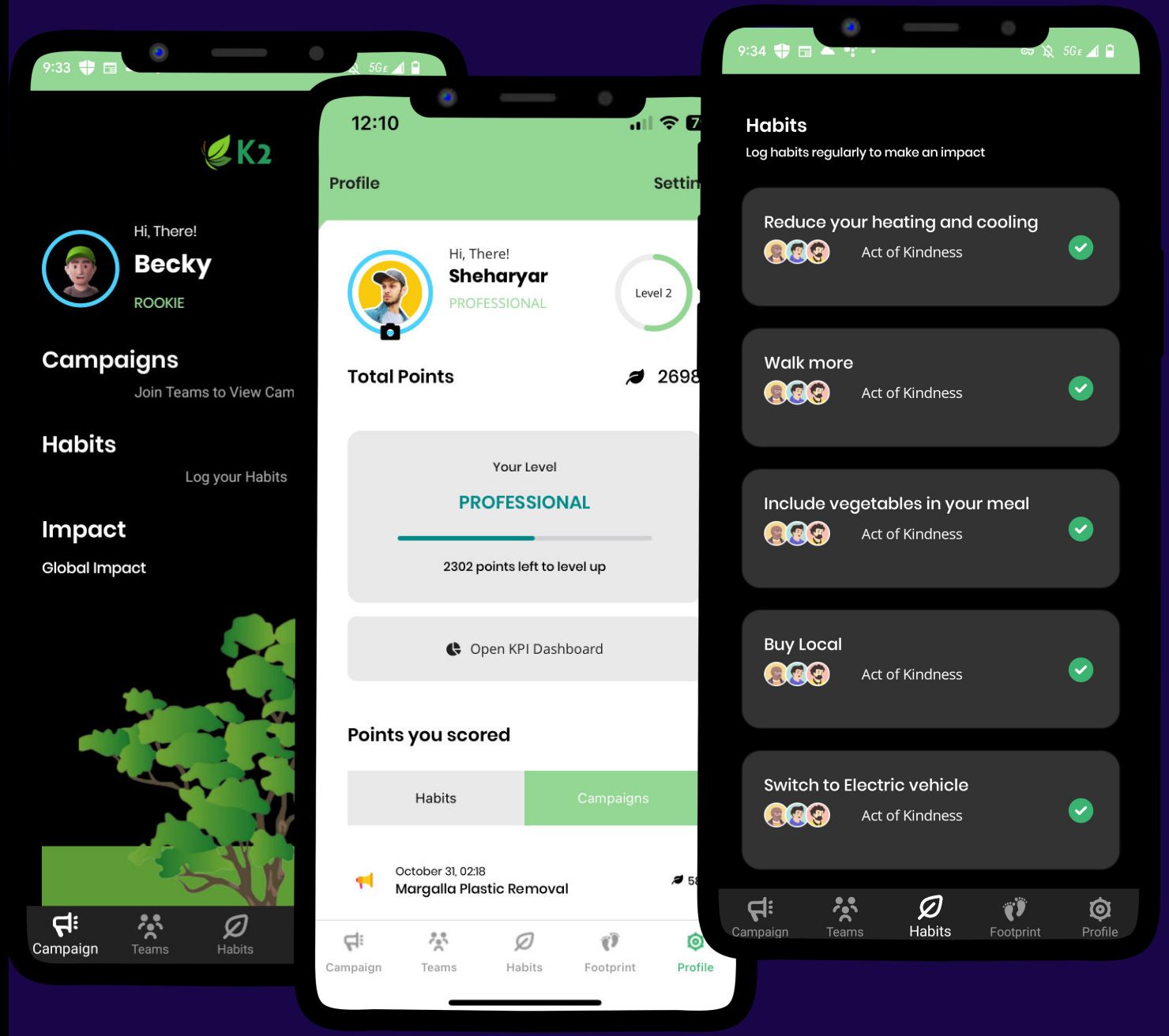
Azure

- App Services
- Cognitive Services
- Notification Hub
- Spatial Anchors



K2 | Klima Kampagne

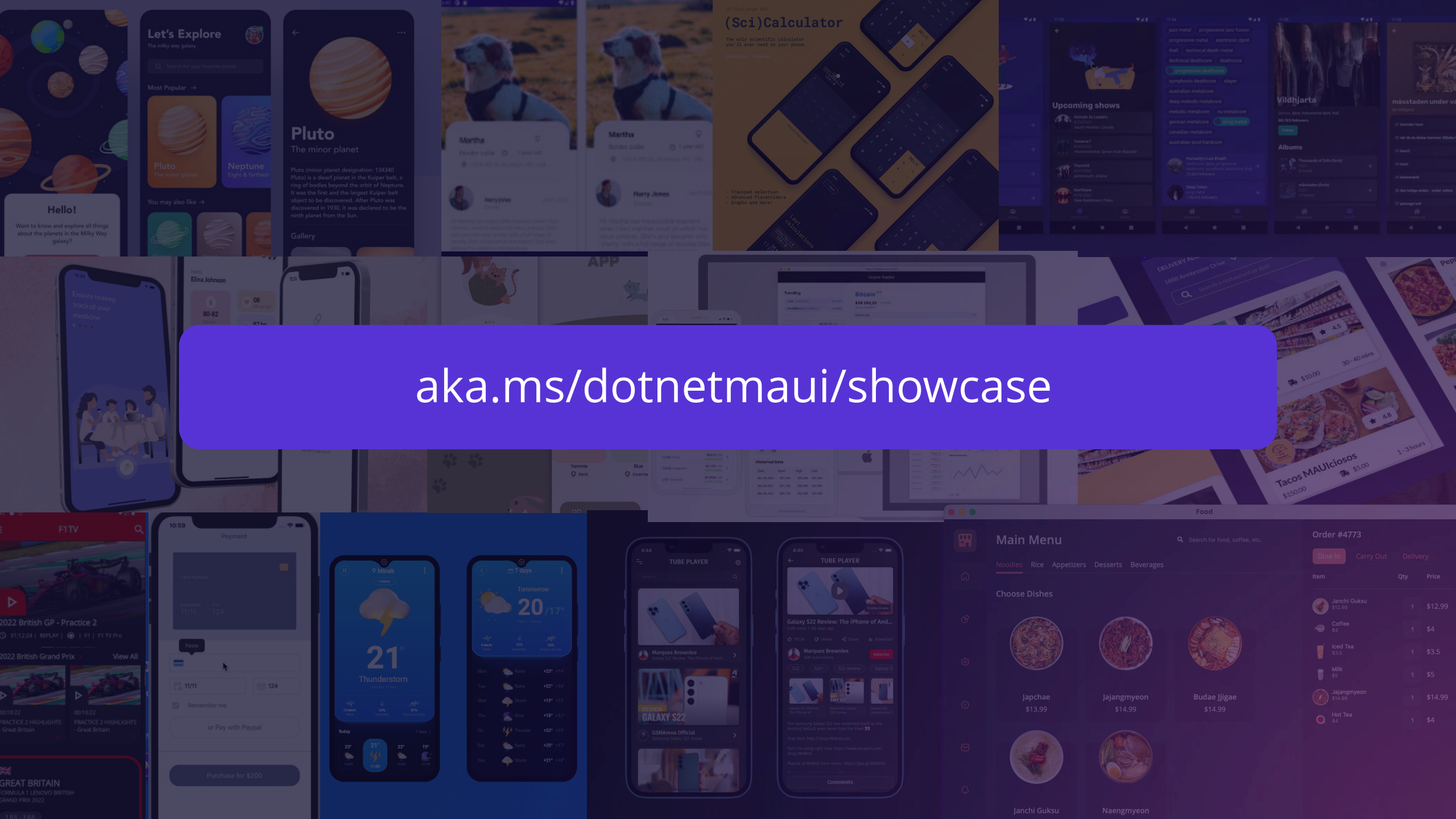
K2 is developed to track climate contributions by individual and large teams/organizations. A campaign management tool to create teams, campaigns and custom habits. Earn points and calculate how much an individual emitted CO2 when taking flight from one city to another.



Logius

Your DigiD allows you to identify yourself when making arrangements on the internet, such as with the government, educational institutes, healthcare institutions or your pension fund.

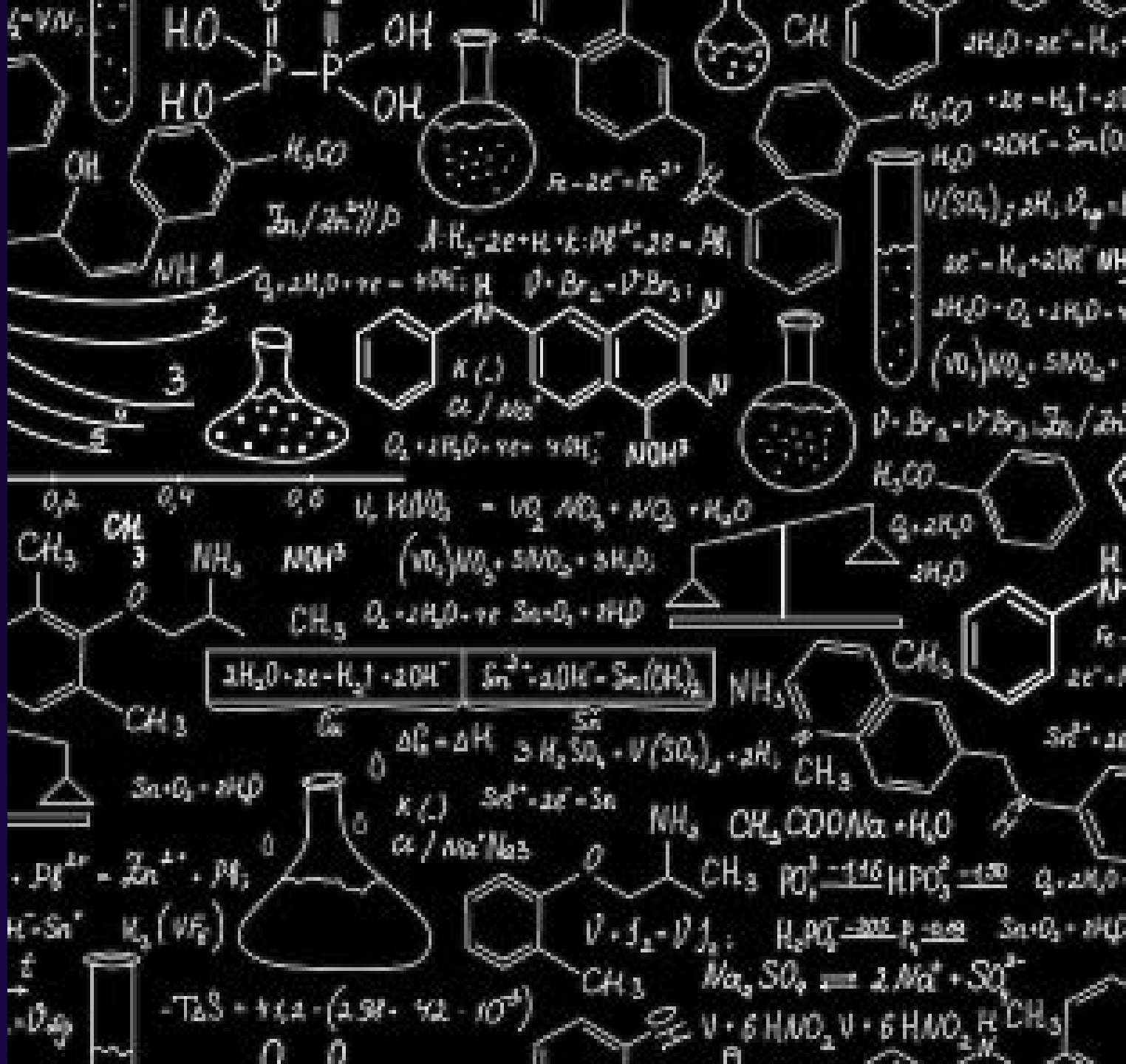




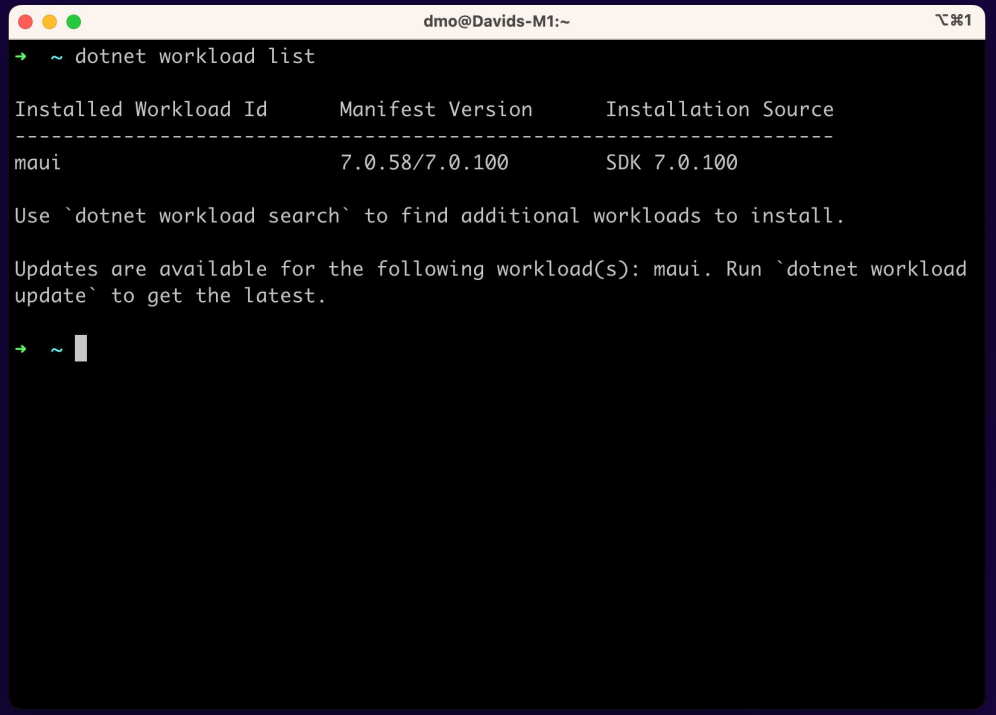
aka.ms/dotnetmaui/showcase

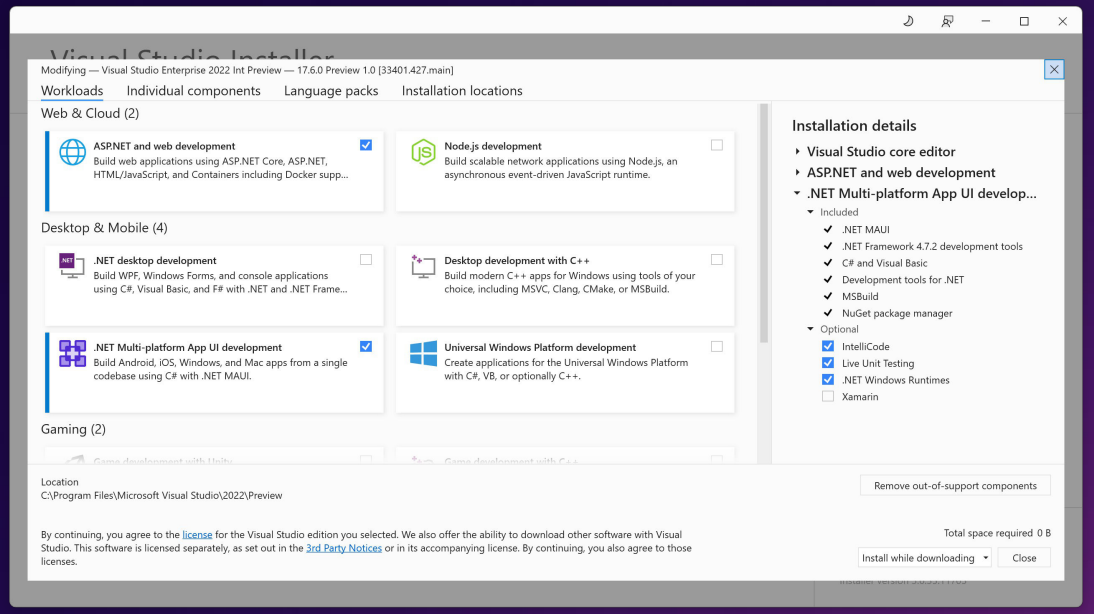
Agenda

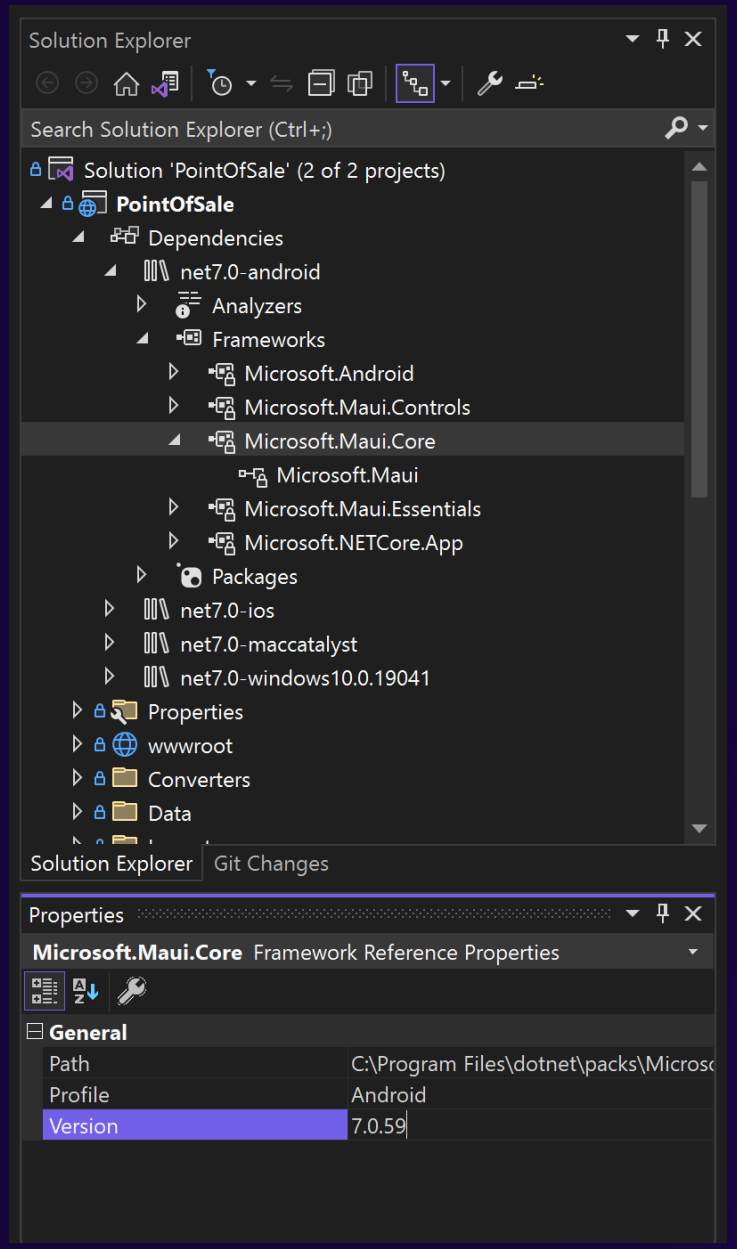
- Workloads
- Upgrading from Xamarin
- Layouts
- Single Project
- Control customization
- Hot Reload

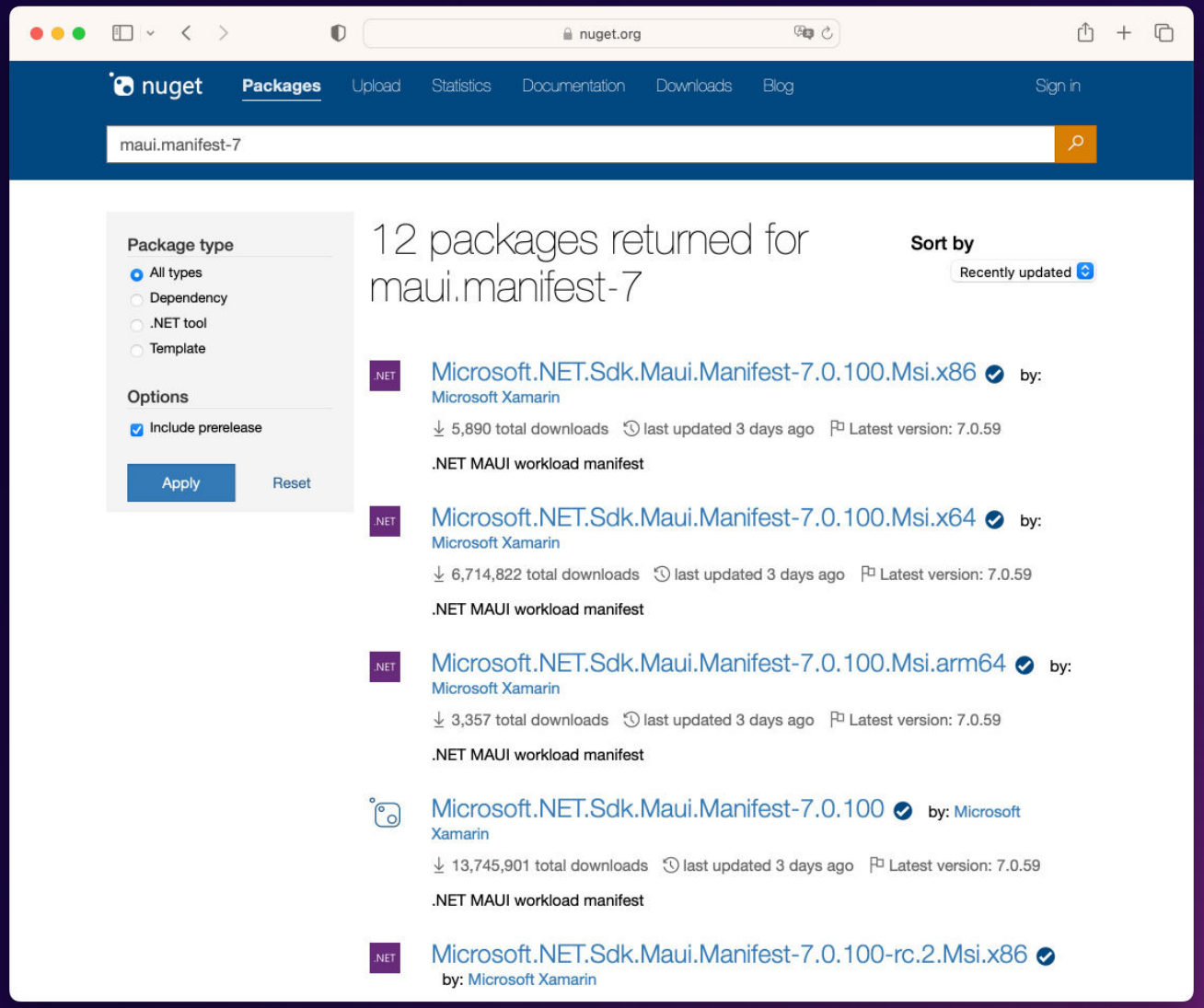


Workloads


dmo@Davids-M1:~
⌥⌘1
➜ ~ dotnet workload list
Installed Workload Id Manifest Version Installation Source
maui 7.0.58/7.0.100 SDK 7.0.100
Use `dotnet workload search` to find additional workloads to install.
Updates are available for the following workload(s): maui. Run `dotnet workload update` to get the latest.
➜ ~


Modifying — Visual Studio Enterprise 2022 Int Preview — 17.6.0 Preview 1.0 [33401.427.main]
Workloads
Individual components
Language packs
Installation locations
Web & Cloud (2)
ASP.NET and web development
Build web applications using ASP.NET Core, ASP.NET, HTML/JavaScript, and Containers including Docker supp...
Node.js development
Build scalable network applications using Node.js, an asynchronous event-driven JavaScript runtime.
Desktop & Mobile (4)
.NET desktop development
Build WPF, Windows Forms, and console applications using C#, Visual Basic, and F# with .NET and .NET Frame...
Desktop development with C++
Build modern C++ apps for Windows using tools of your choice, including MSVC, Clang, CMake, or MSBuild.
.NET Multi-platform App UI development
Build Android, iOS, Windows, and Mac apps from a single codebase using C# with .NET MAUI.
Universal Windows Platform development
Create applications for the Universal Windows Platform with C#, VB, or optionally C++.
Gaming (2)
Installation details
Visual Studio core editor
ASP.NET and web development
.NET Multi-platform App UI develop...
Included
.NET MAUI
.NET Framework 4.7.2 development tools
C# and Visual Basic
Development tools for .NET
MSBuild
NuGet package manager
Optional
IntelliCode
Live Unit Testing
.NET Windows Runtimes
Xamarin
Location
C:\Program Files\Microsoft Visual Studio\2022\Preview
Remove out-of-support components
By continuing, you agree to the license for the Visual Studio edition you selected. We also offer the ability to download other software with Visual Studio. This software is licensed separately, as set out in the 3rd Party Notices or in its accompanying license. By continuing, you also agree to those licenses.
Total space required 0 B
Install while downloading
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Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'PointOfSale' (2 of 2 projects)
PointOfSale
Dependencies
net7.0-android
Analyzers
Frameworks
Microsoft.Android
Microsoft.Maui.Controls
Microsoft.Maui.Core
Microsoft.Maui
Microsoft.Maui.Essentials
Microsoft.NETCore.App
Packages
net7.0-ios
net7.0-maccatalyst
net7.0-windows10.0.19041
Properties
wwwroot
Converters
Data
Solution Explorer
Git Changes
Properties
Microsoft.Maui.Core Framework Reference Properties
General
Path
C:\Program Files\dotnet\packs\Microsc
Profile
Android
Version
7.0.59


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maui.manifest-7
Package type
All types
Dependency
.NET tool
Template
Options
Include prerelease
Apply
Reset
12 packages returned for maui.manifest-7
Sort by
Recently updated
Microsoft.NET.Sdk.Maui.Manifest-7.0.100.Msi.x86 by: Microsoft Xamarin
5,890 total downloads
last updated 3 days ago
Latest version: 7.0.59
.NET MAUI workload manifest
Microsoft.NET.Sdk.Maui.Manifest-7.0.100.Msi.x64 by: Microsoft Xamarin
6,714,822 total downloads
last updated 3 days ago
Latest version: 7.0.59
.NET MAUI workload manifest
Microsoft.NET.Sdk.Maui.Manifest-7.0.100.Msi.arm64 by: Microsoft Xamarin
3,357 total downloads
last updated 3 days ago
Latest version: 7.0.59
.NET MAUI workload manifest
Microsoft.NET.Sdk.Maui.Manifest-7.0.100 by: Microsoft Xamarin
13,745,901 total downloads
last updated 3 days ago
Latest version: 7.0.59
.NET MAUI workload manifest
Microsoft.NET.Sdk.Maui.Manifest-7.0.100-rc.2.Msi.x86
by: Microsoft Xamarin

dotnet.microsoft.com

Download .NET 7.0

Not sure what to download? See recommended downloads for the latest version of .NET.

7.0.2

Release notes **Latest release date** January 10, 2023

Build apps - SDK

SDK 7.0.102

OS	Installers	Binaries
Linux	Package manager instructions	Arm32 | Arm32 Alpine | Arm64 | Arm64 Alpine | x64 | x64 Alpine
macOS	Arm64 | x64	Arm64 | x64
Windows	Arm64 | x64 | x86 | winget instructions	Arm64 | x64 | x86
All	dotnet-install scripts	

Visual Studio support
Visual Studio 2022 (v17.4)
Visual Studio 2022 for Mac (v17.4)

Included in
Visual Studio 17.4.4

Run apps - Runtime

ASP.NET Core Runtime 7.0.2

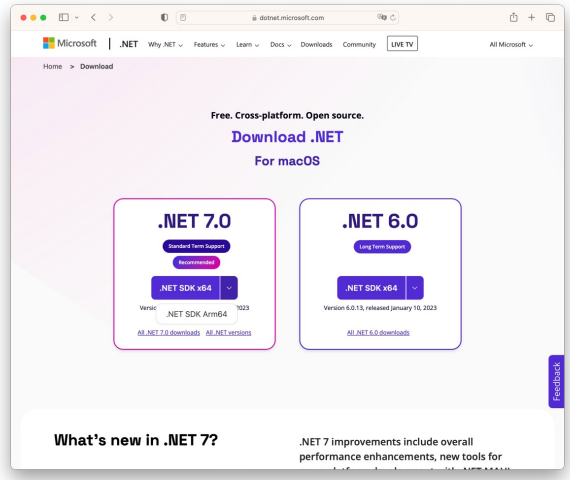
The ASP.NET Core Runtime enables you to run existing web/server applications. **On Windows, we recommend installing the Hosting Bundle, which includes the .NET Runtime and IIS support.**

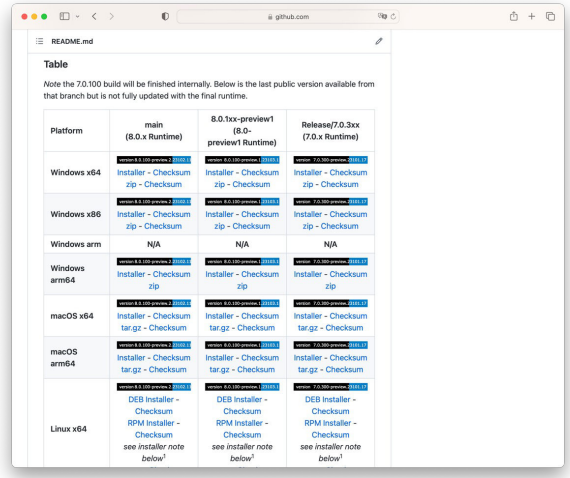
IIS runtime support (ASP.NET Core Module v2)
17.0.22341.2

OS	Installers	Binaries
Linux	Package manager instructions	Arm32 | Arm32 Alpine | Arm64 | Arm64 Alpine | x64 | x64 Alpine
macOS		Arm64 | x64
Windows	Hosting Bundle | x64 | x86 | winget instructions	Arm64 | x64 | x86

.NET Desktop Runtime 7.0.2

Feedback





Commands

> dotnet sdk check

> dotnet –version

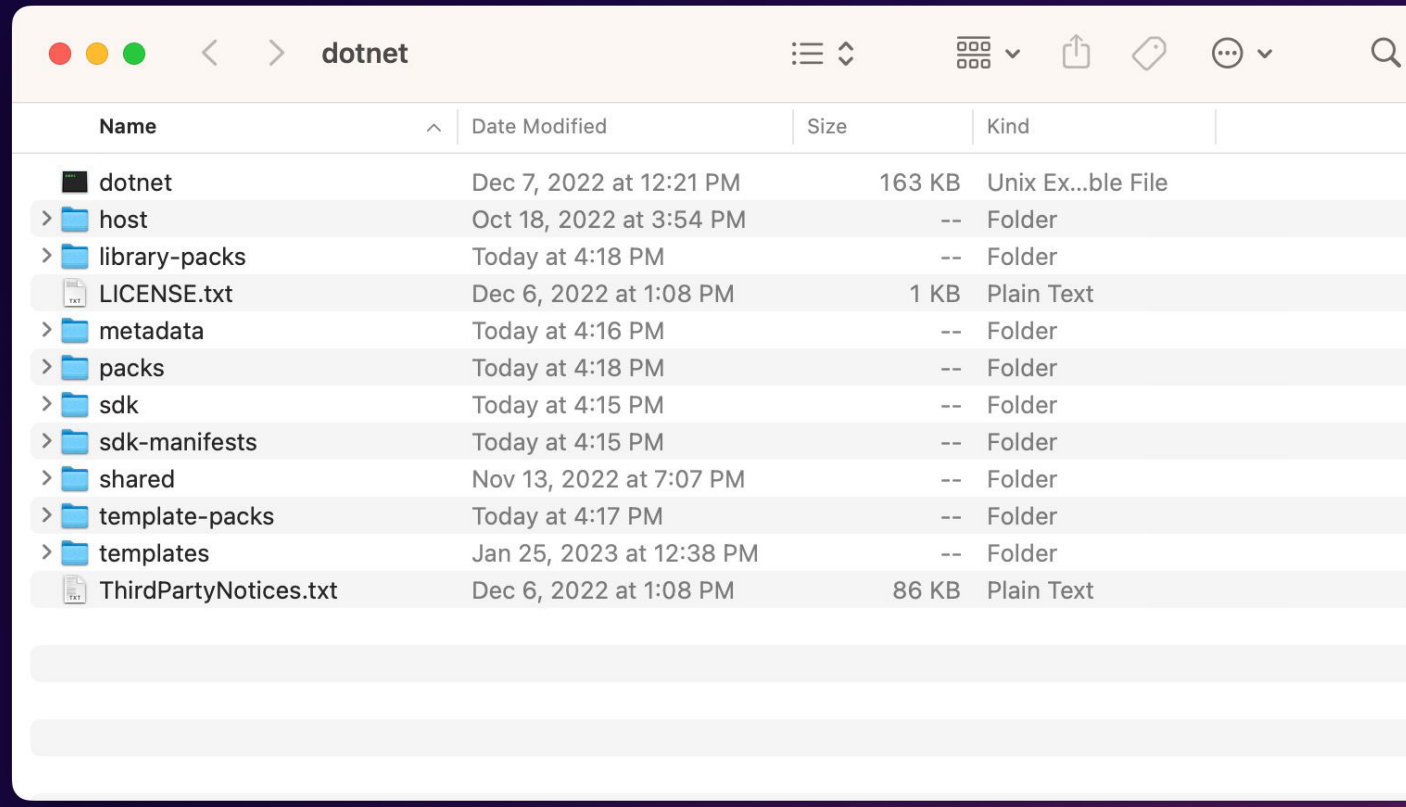
> dotnet workload list

> dotnet workload install maui (sudo)

> dotnet workload update (sudo)

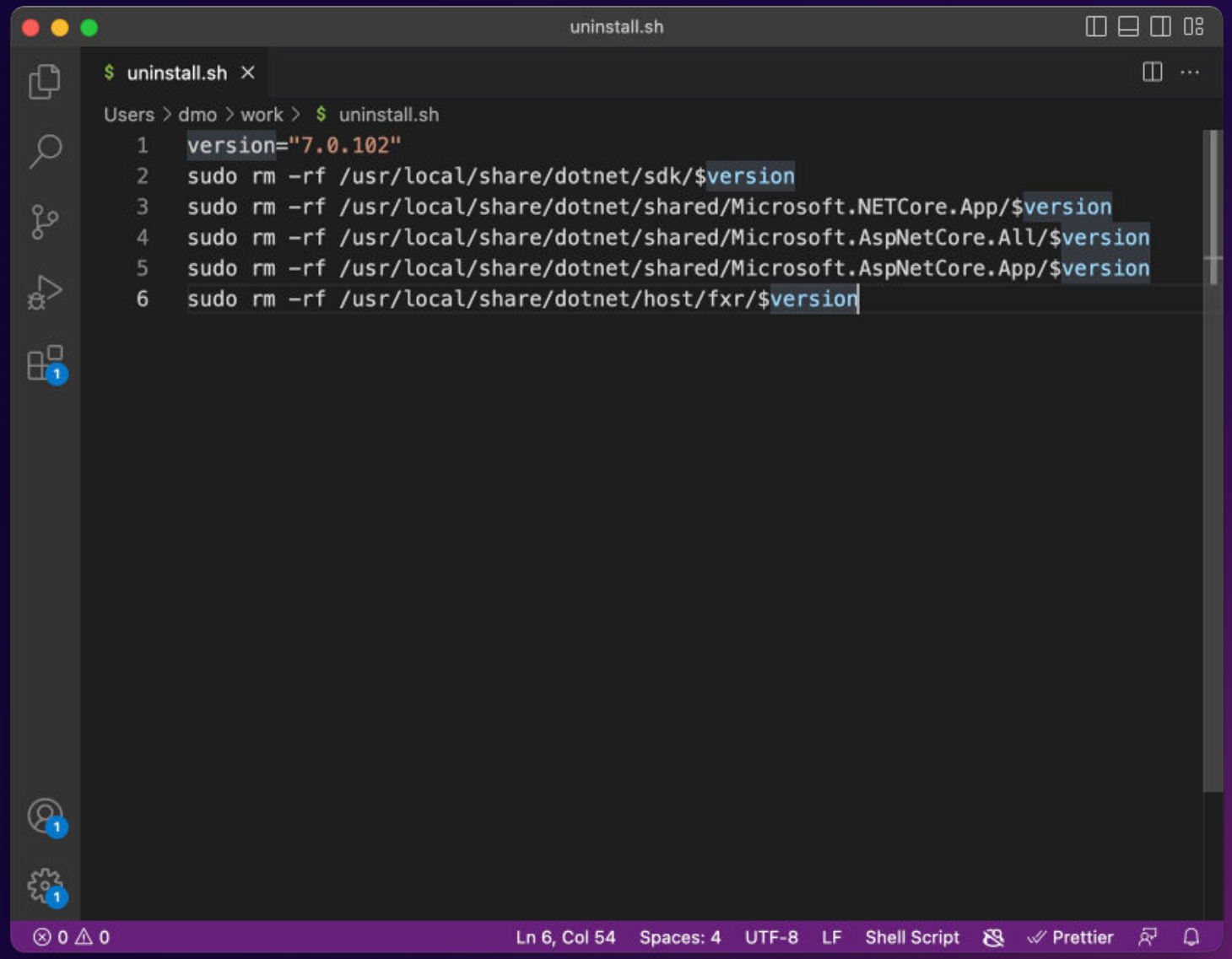
> dotnet nuget list source

> which dotnet



uninstall.sh

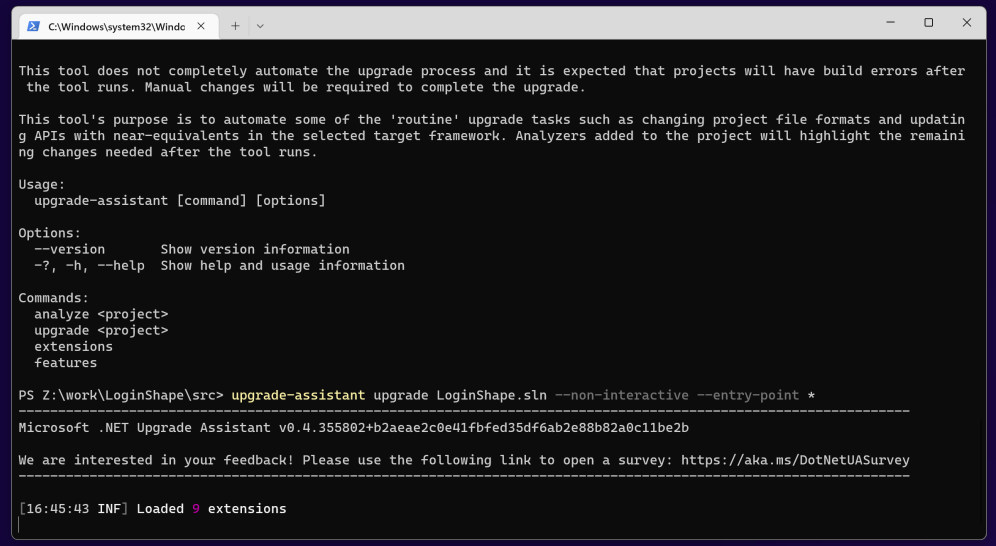
> ./uninstall.sh



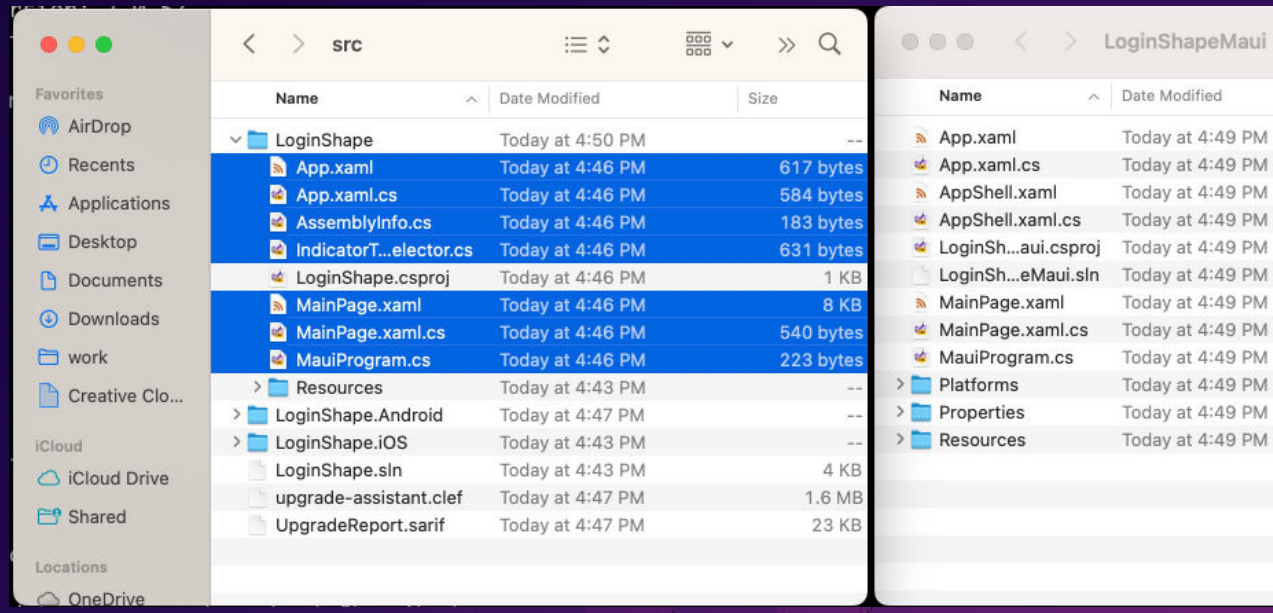
Upgrading from Xamarin

Two Options

- .NET Upgrade Assistant

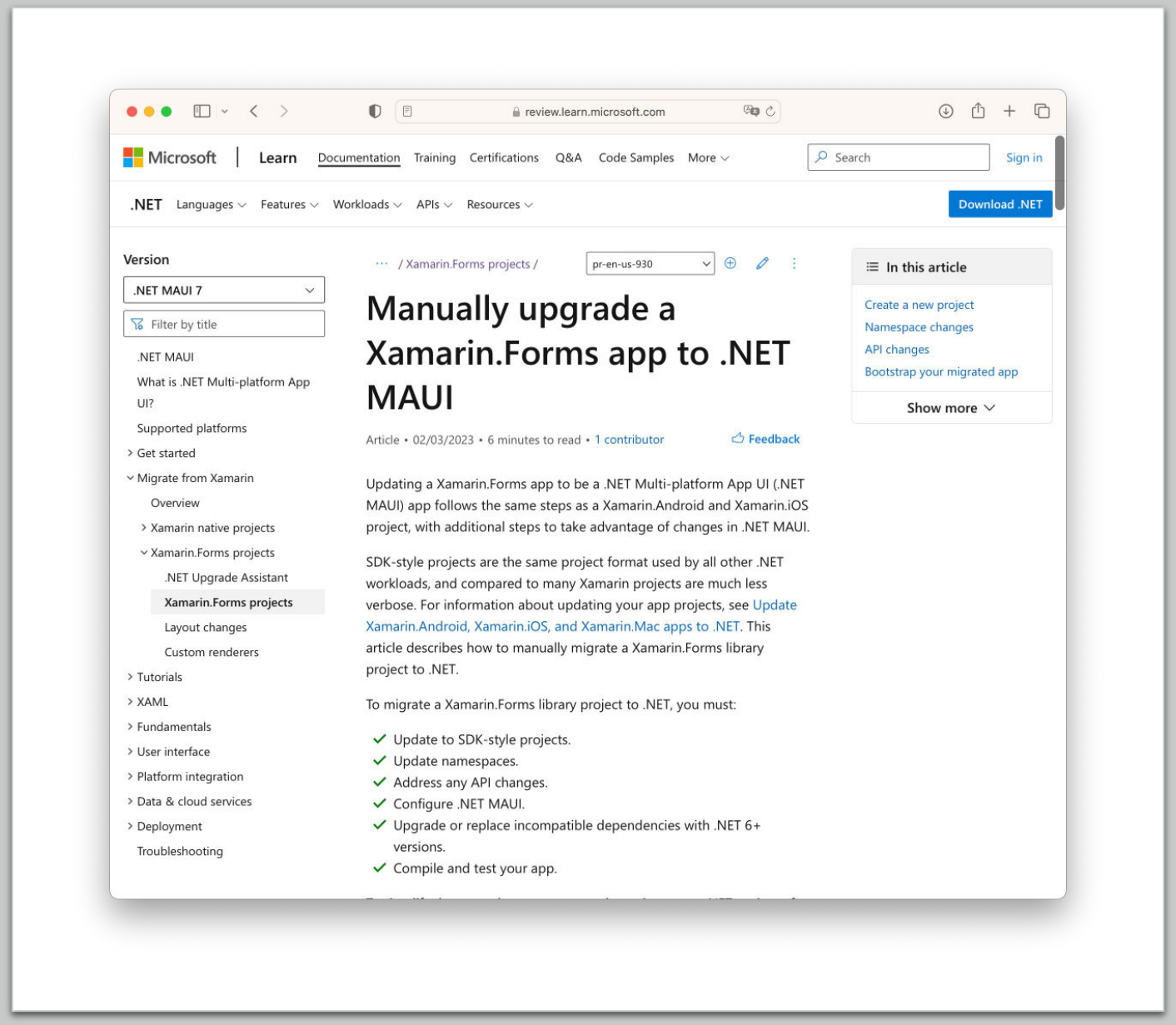


- Copy Pasta

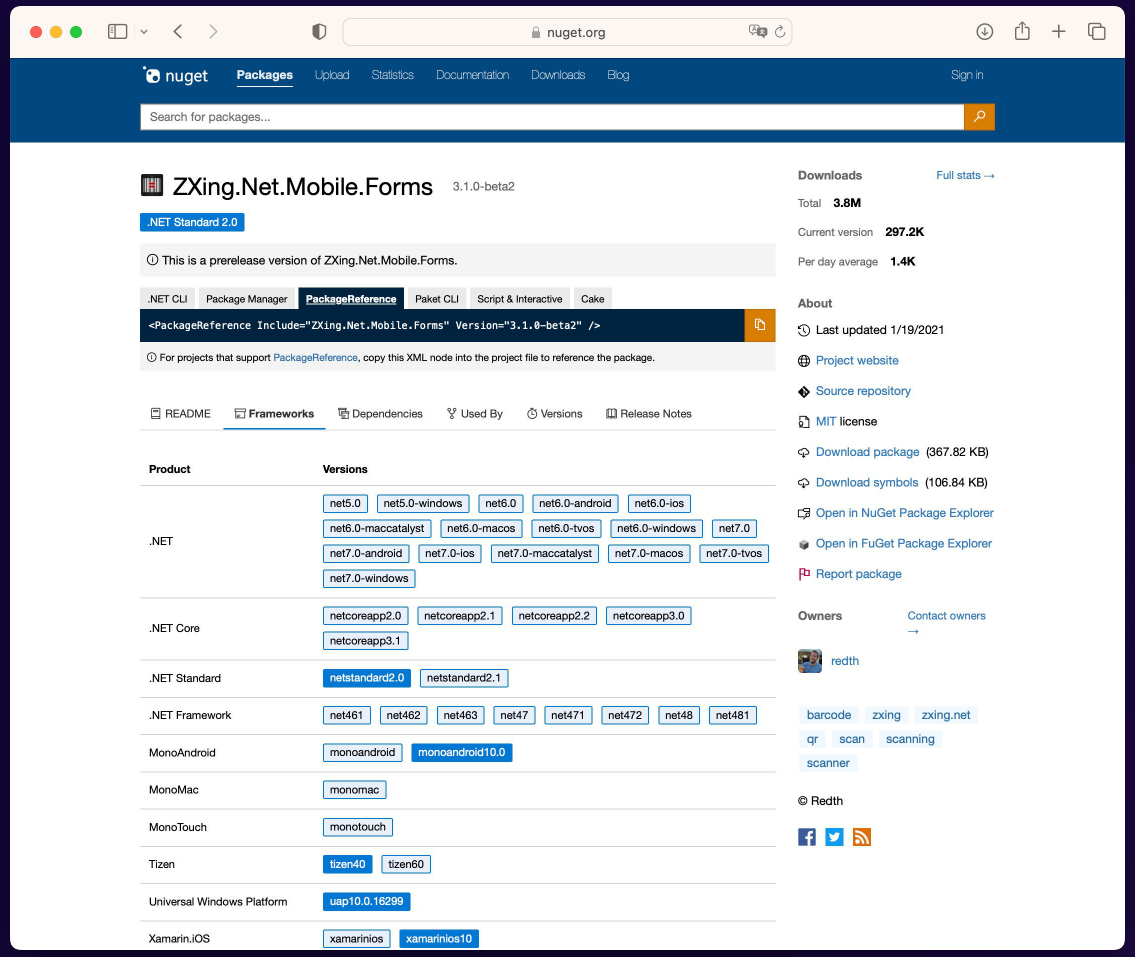


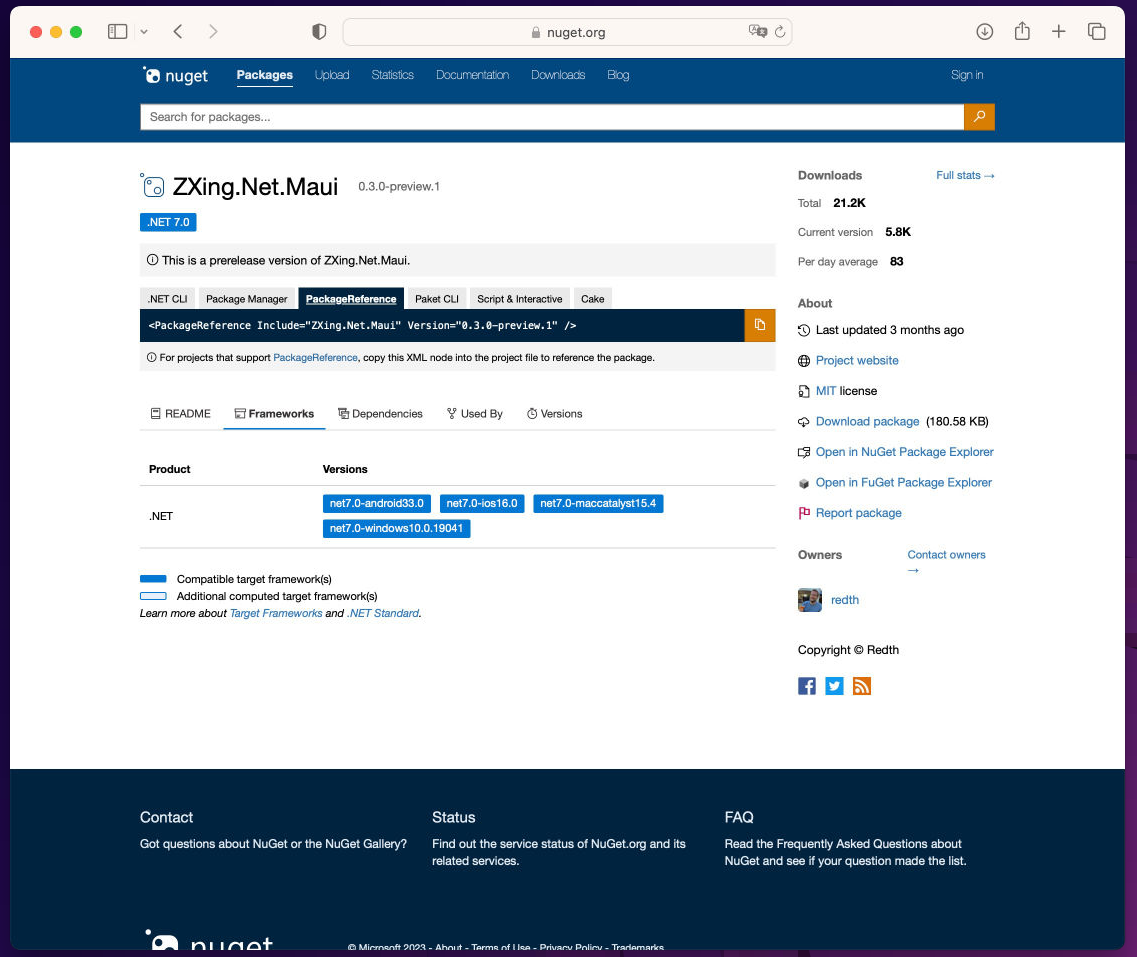
Upgrade tasks

- Update to SDK-style projects.
- Update namespaces.
- Address any API changes.
- Configure .NET MAUI.
- Upgrade or replace incompatible dependencies with .NET 6+ versions.
- Compile and test your app.

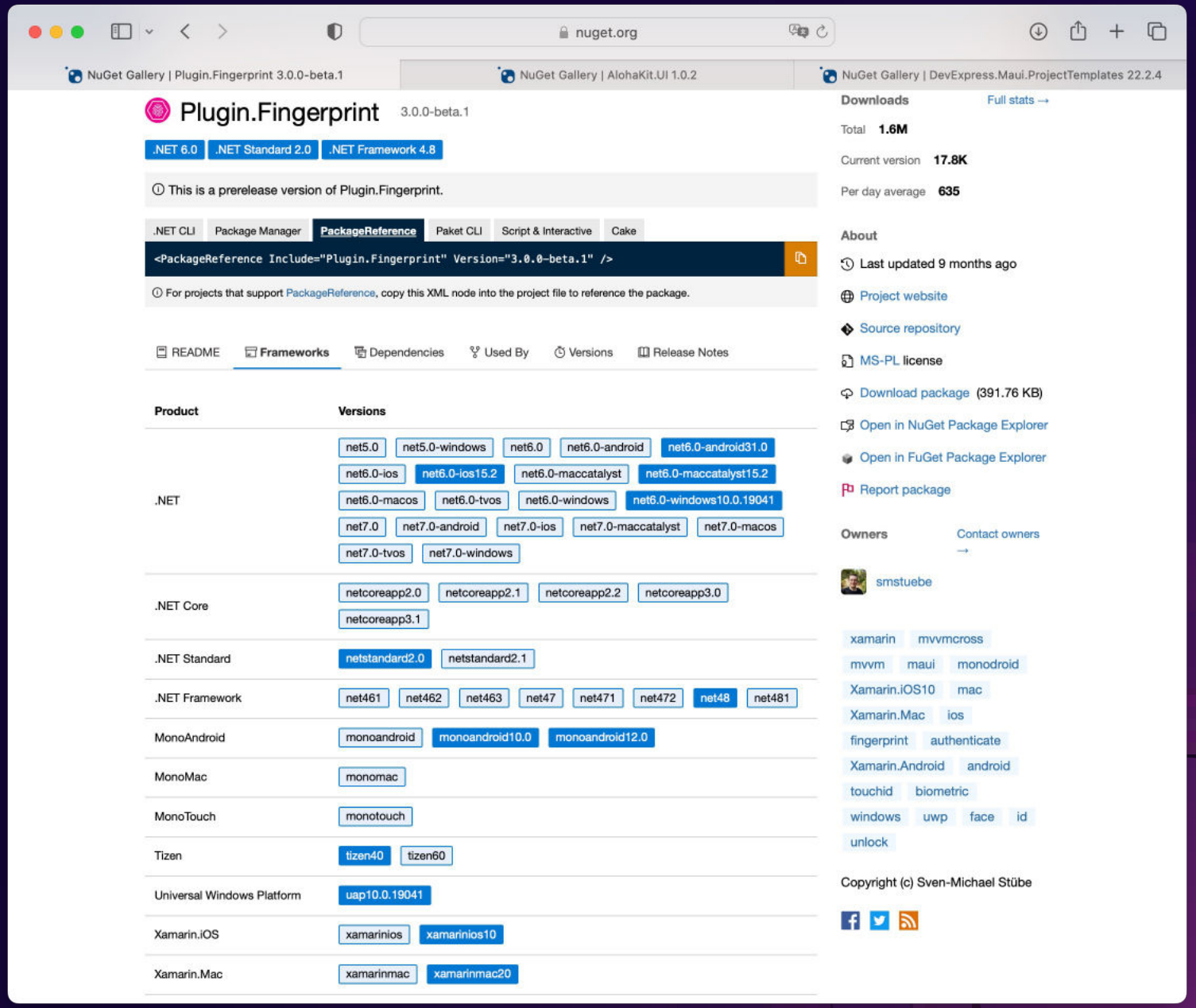


Which Nuget?

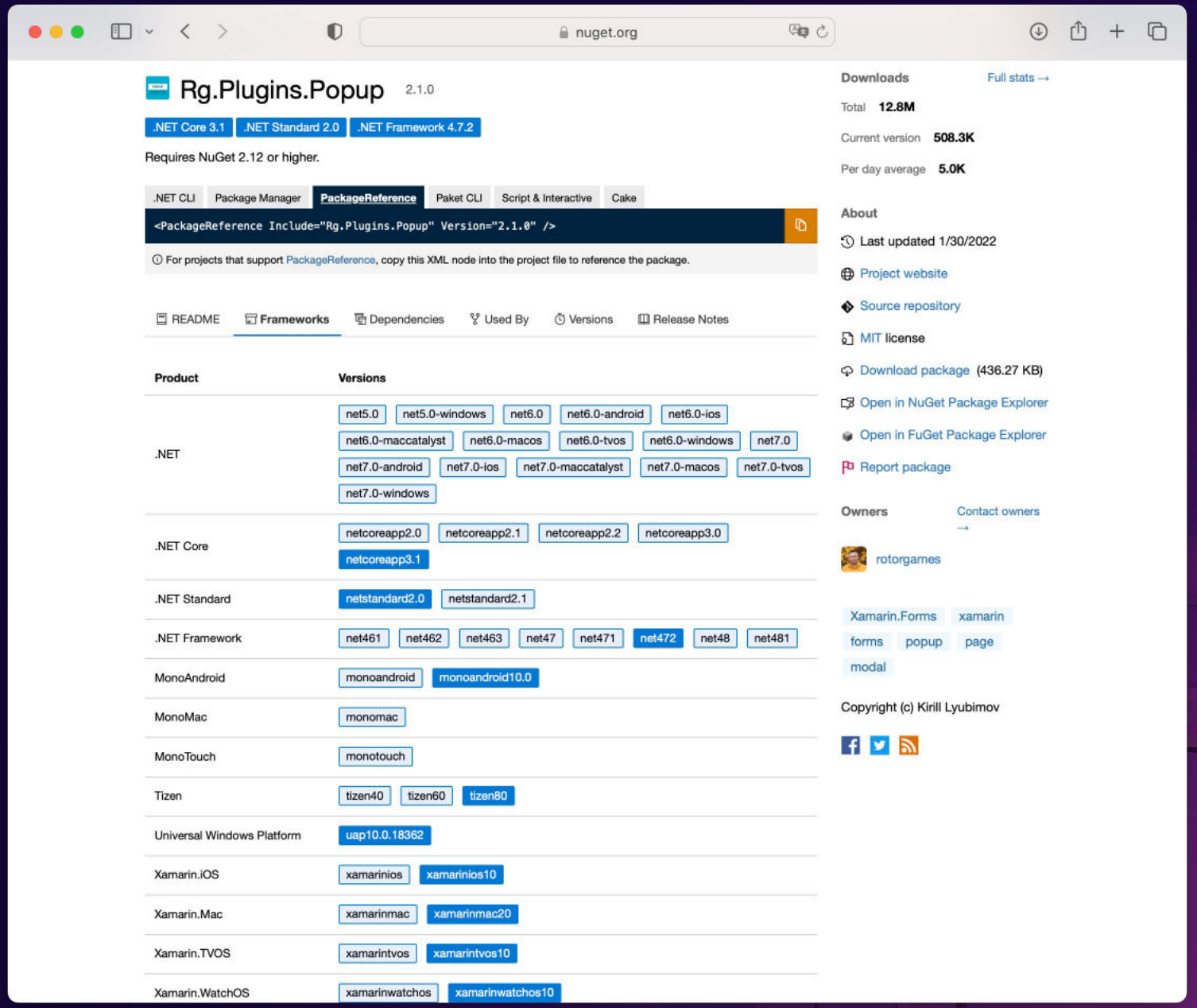

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ZXing.Net.Mobile.Forms
3.1.0-beta2
.NET Standard 2.0
This is a prerelease version of ZXing.Net.Mobile.Forms.
.NET CLI
Package Manager
PackageReference
Paket CLI
Script & Interactive
Cake
<PackageReference Include="ZXing.Net.Mobile.Forms" Version="3.1.0-beta2" />
For projects that support PackageReference, copy this XML node into the project file to reference the package.
README
Frameworks
Dependencies
Used By
Versions
Release Notes
Product
Versions
.NET
net5.0 net5.0-windows net6.0 net6.0-android net6.0-ios net6.0-maccatalyst net6.0-macos net6.0-tvos net6.0-windows net7.0 net7.0-android net7.0-ios net7.0-maccatalyst net7.0-macos net7.0-tvos net7.0-windows
.NET Core
netcoreapp2.0 netcoreapp2.1 netcoreapp2.2 netcoreapp3.0 netcoreapp3.1
.NET Standard
netstandard2.0 netstandard2.1
.NET Framework
net461 net462 net463 net47 net471 net472 net48 net481
MonoAndroid
monoandroid monoandroid10.0
MonoMac
monomac
MonoTouch
monotouch
Tizen
tizen40 tizen60
Universal Windows Platform
uap10.0.16299
Xamarin.iOS
xamarinios xamarinios10
Downloads
Full stats →
Total 3.8M
Current version 297.2K
Per day average 1.4K
About
Last updated 1/19/2021
Project website
Source repository
MIT license
Download package (367.82 KB)
Download symbols (106.84 KB)
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Open in FuGet Package Explorer
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This is a prerelease version of ZXing.Net.Maui.
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Package Manager
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net7.0-android33.0 net7.0-ios16.0 net7.0-maccatalyst15.4 net7.0-windows10.0.19041
Compatible target framework(s)
Additional computed target framework(s)
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Current version 5.8K
Per day average 83
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Project website
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NuGet Gallery | Plugin.Fingerprint 3.0.0-beta.1
NuGet Gallery | AlohaKit.UI 1.0.2
NuGet Gallery | DevExpress.Maui.ProjectTemplates 22.2.4
Plugin.Fingerprint 3.0.0-beta.1
.NET 6.0
.NET Standard 2.0
.NET Framework 4.8
This is a prerelease version of Plugin.Fingerprint.
.NET CLI
Package Manager
PackageReference
Paket CLI
Script & Interactive
Cake
<PackageReference Include="Plugin.Fingerprint" Version="3.0.0-beta.1" />
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.NET Core
netcoreapp2.0 netcoreapp2.1 netcoreapp2.2 netcoreapp3.0 netcoreapp3.1
.NET Standard
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.NET Framework
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MonoMac
monomac
MonoTouch
monotouch
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Xamarin.Mac
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Total 1.6M
Current version 17.8K
Per day average 635
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Last updated 9 months ago
Project website
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Rg.Plugins.Popup 2.1.0
.NET Core 3.1
.NET Standard 2.0
.NET Framework 4.7.2
Requires NuGet 2.12 or higher.
.NET CLI
Package Manager
PackageReference
Paket CLI
Script & Interactive
Cake
<PackageReference Include="Rg.Plugins.Popup" Version="2.1.0" />
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.NET Core
netcoreapp2.0 netcoreapp2.1 netcoreapp2.2 netcoreapp3.0 netcoreapp3.1
.NET Standard
netstandard2.0 netstandard2.1
.NET Framework
net461 net462 net463 net47 net471 net472 net48 net481
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monotouch
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Xamarin.Forms xamarin forms popup page modal
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Layouts

Behavior changes

Layout	Xamarin.Forms	.NET MAUI	Recommendation
All	In certain cases sizing requests would not be honored.	Sizing requests are honored.	
Grid	Columns and rows can be inferred from XAML.	Columns and rows must be explicitly declared.	Add ColumnDefinitions and RowDefinitions.
HorizontalStackLayout		"*AndExpand" has no effect.	
RelativeLayout		Requires the compatibility namespace.	Use Grid.
ScrollView	Arbitrary minimum size, inconsistent sizing	Expands to fill all space	Set a size if within a stack to enable scrolling.
StackLayout	Children can fill space in the stacking direction.	Children are stacked and will go beyond available space.	If you need child views to fill space, change to a Grid.
VerticalStackLayout		"*AndExpand" has no effect.	

Value changes

Property	Xamarin.Forms value	.NET MAUI value
Grid.ColumnSpacing	6	0
Grid.RowSpacing	6	0
StackLayout.Spacing	6	0

StackLayout

- Very simple – stacks in a single direction until all children stacked.
- Does NOT subdivide space
- Remove any "AndExpand" properties which are orthogonal to the stacking direction.

XAML Copy

```xaml
<StackLayout>
    <Label Text="Hello world!"/>
    <Image VerticalOptions="FillAndExpand" Source="dotnetbot.png"/>
</StackLayout>
```

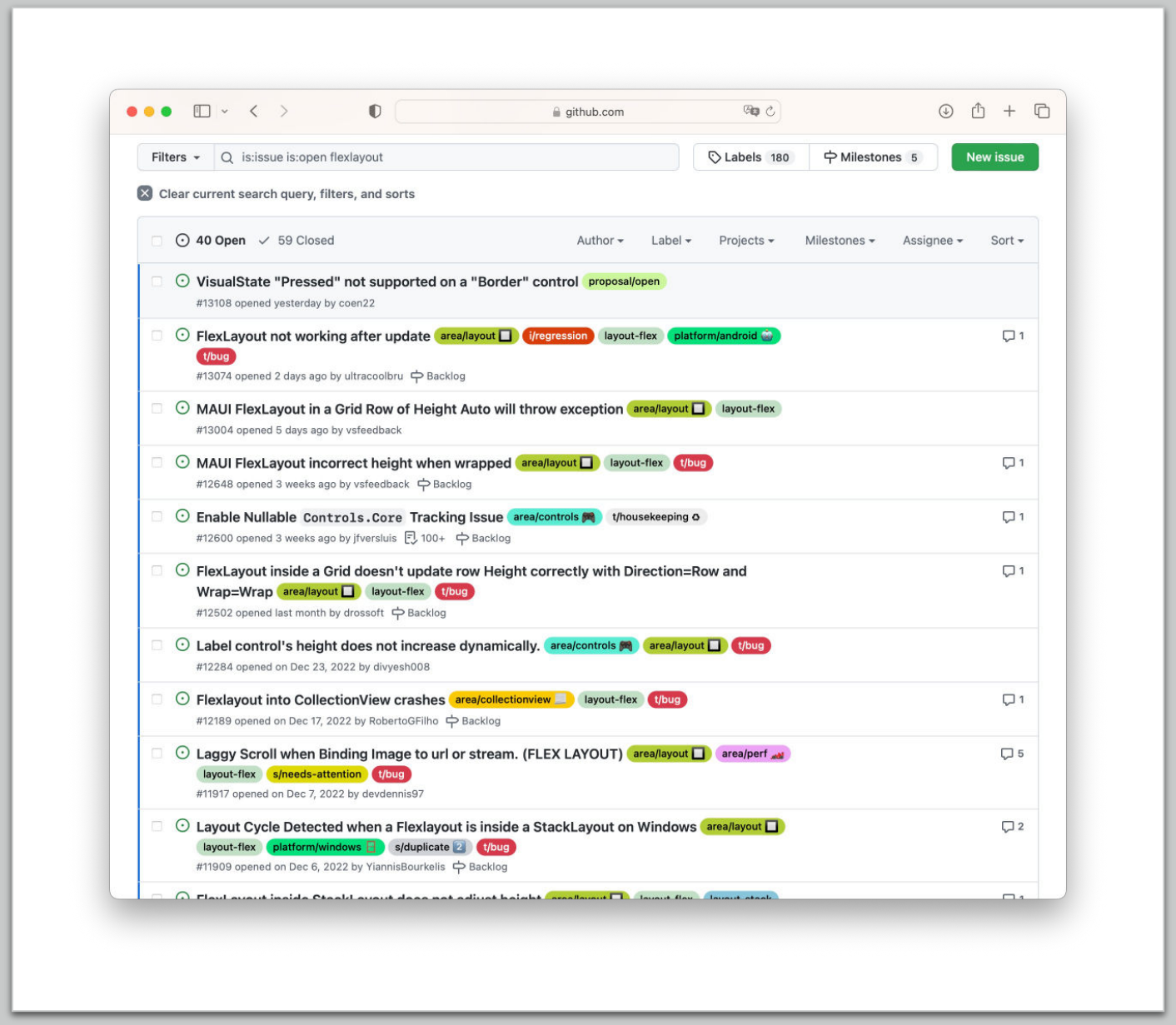
This can be converted to a `Grid` in .NET MAUI:

XAML Copy

```xaml
<Grid RowDefinitions="Auto, *">
    <Label Text="Hello world!"/>
    <Image Grid.Row="1" Source="dotnetbot.png"/>
</StackLayout>
```

FlexLayout

- Be careful



Custom layouts

- Create an ILayoutManager and override:
  - Measure
  - ArrangeChildren

https://github.com/hartez/CustomLayoutExamples

So given this XAML:

```
<local:HorizontalWrapLayout Margin="10" Spacing="5">
    <Button Text="One"></Button>
    <Button Text="Two"></Button>
    <Button Text="Three"></Button>
    <Button Text="Four"></Button>
    <Button Text="Five"></Button>
    <Button Text="Six"></Button>
    <Button Text="Seven"></Button>
    <Button Text="Eight"></Button>
    <Button Text="Nine"></Button>
    <Button Text="Ten"></Button>
</local:HorizontalWrapLayout>
```

You'll see something like this:

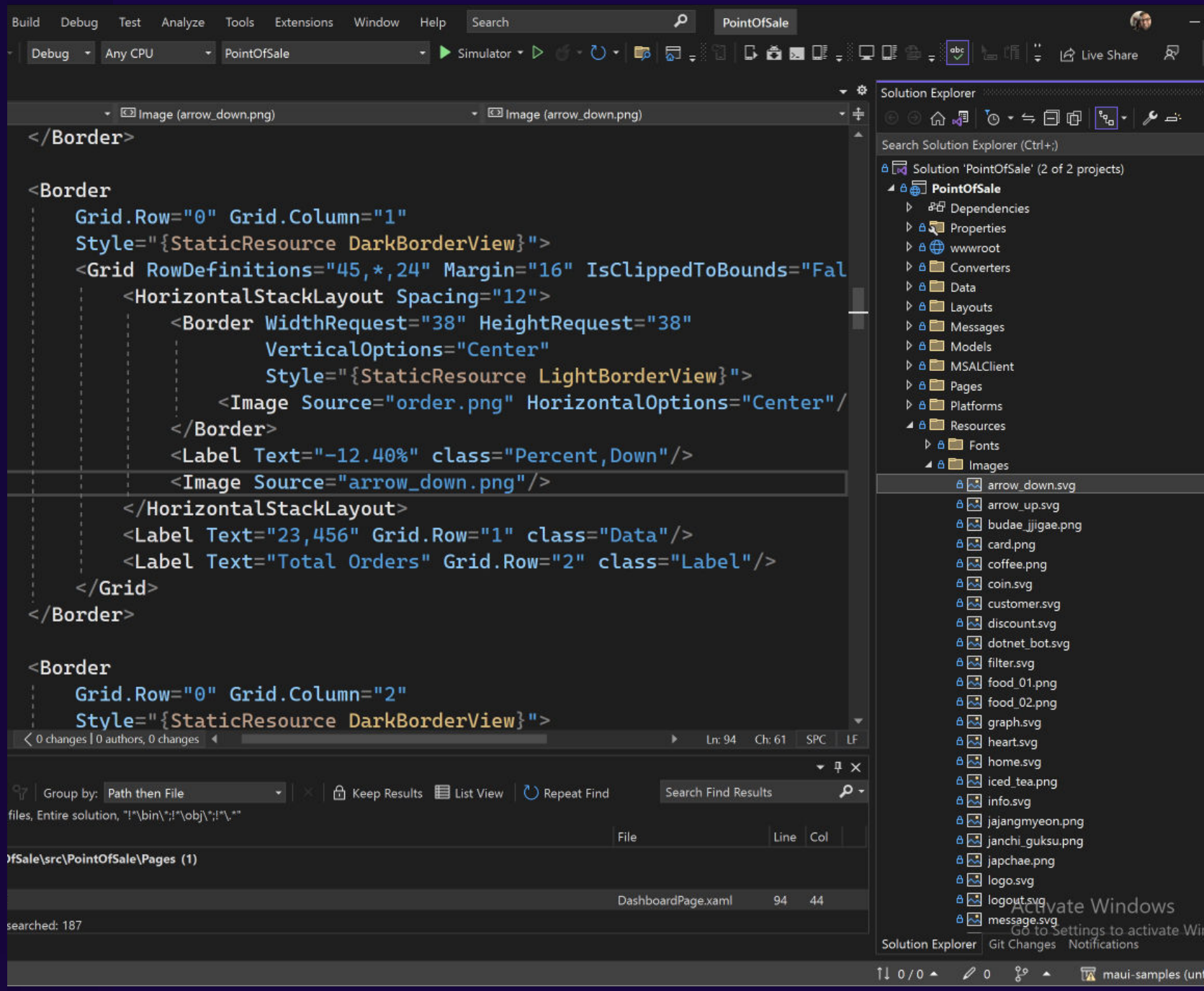


Single Project

Images

- SVGs generate PNGs
- Using the SVG extension still renders a PNG
- Edit the csproj to set:
  - BaseSize
  - Resize
  - Color
  - TintColor

Filenames must be lowercase, start and end with a letter character, and contain only alphanumeric characters or underscores.

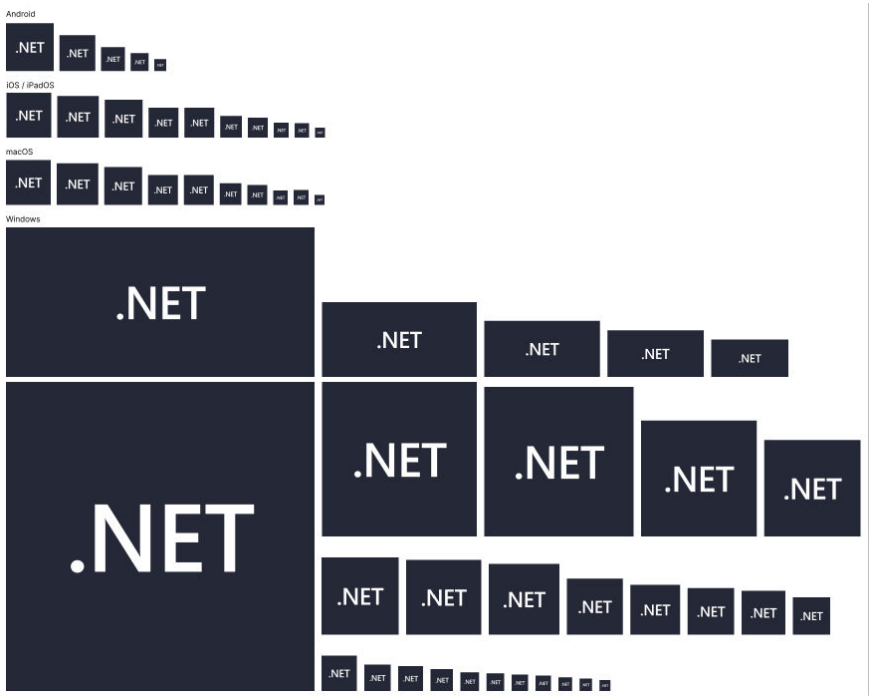


App icons and Splash screens

- .NET MAUI makes this easy with :
  - MauiIcon
  - MauiSplashScreen

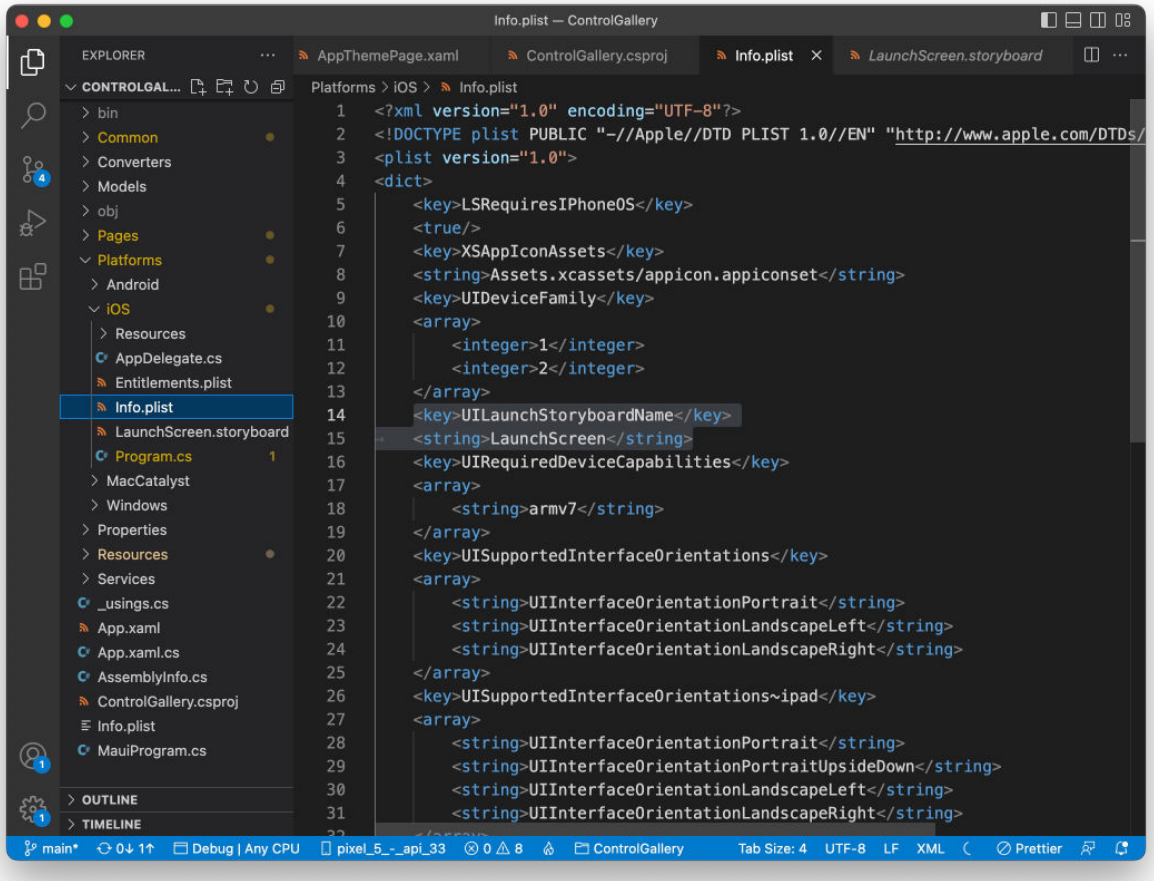
```
<ItemGroup>
    <!-- App Icon -->
    <MauiIcon
        Include="Resources\AppIcon\appicon.svg"
        ForegroundFile="Resources\AppIcon\appiconfg.svg"
        Color="#512BD4" />

    <!-- Splash Screen -->
    <MauiSplashScreen
        Include="Resources\Splash\splash.svg"
        Color="#512BD4"
        BaseSize="128,128" />
```



App icons and Splash screens

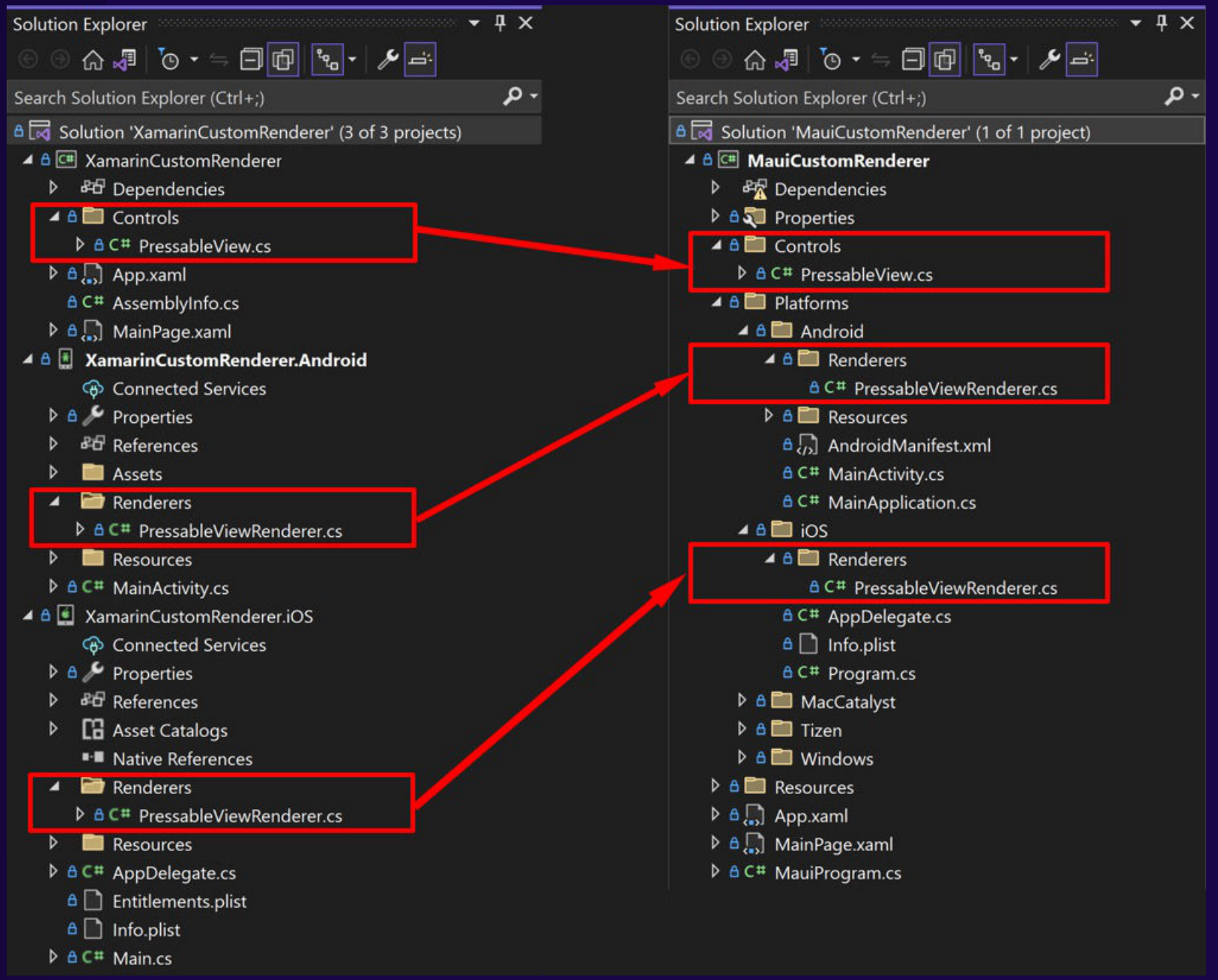
- But you can still make your own “by hand”



Control Customization

Custom Renderers

- Platform folders will multi-target appropriately
- This is optional



Custom Renderers

- Remove [ExportRenderer]
- Configure as a handler
- Use it

C# Copy

```
using Microsoft.Maui.Controls.Compatibility.Hosting;
...

public static class MauiProgram
{
    public static MauiApp CreateMauiApp()
    {
        var builder = MauiApp.CreateBuilder();
        builder
            .UseMauiApp<App>()
            .UseMauiCompatibility()
            .ConfigureFonts(fonts =>
            {
                fonts.AddFont("OpenSans-Regular.ttf", "OpenSansRegular");
                fonts.AddFont("OpenSans-Semibold.ttf", "OpenSansSemibold");
            })
            .ConfigureMauiHandlers((handlers) =>
            {
#if ANDROID
                handlers.AddHandler(typeof(PressableView), typeof(XamarinCustomRend
#elif IOS
                handlers.AddHandler(typeof(PressableView), typeof(XamarinCustomRend
#endif
            });

        return builder.Build();
    }
}
```

Hooks

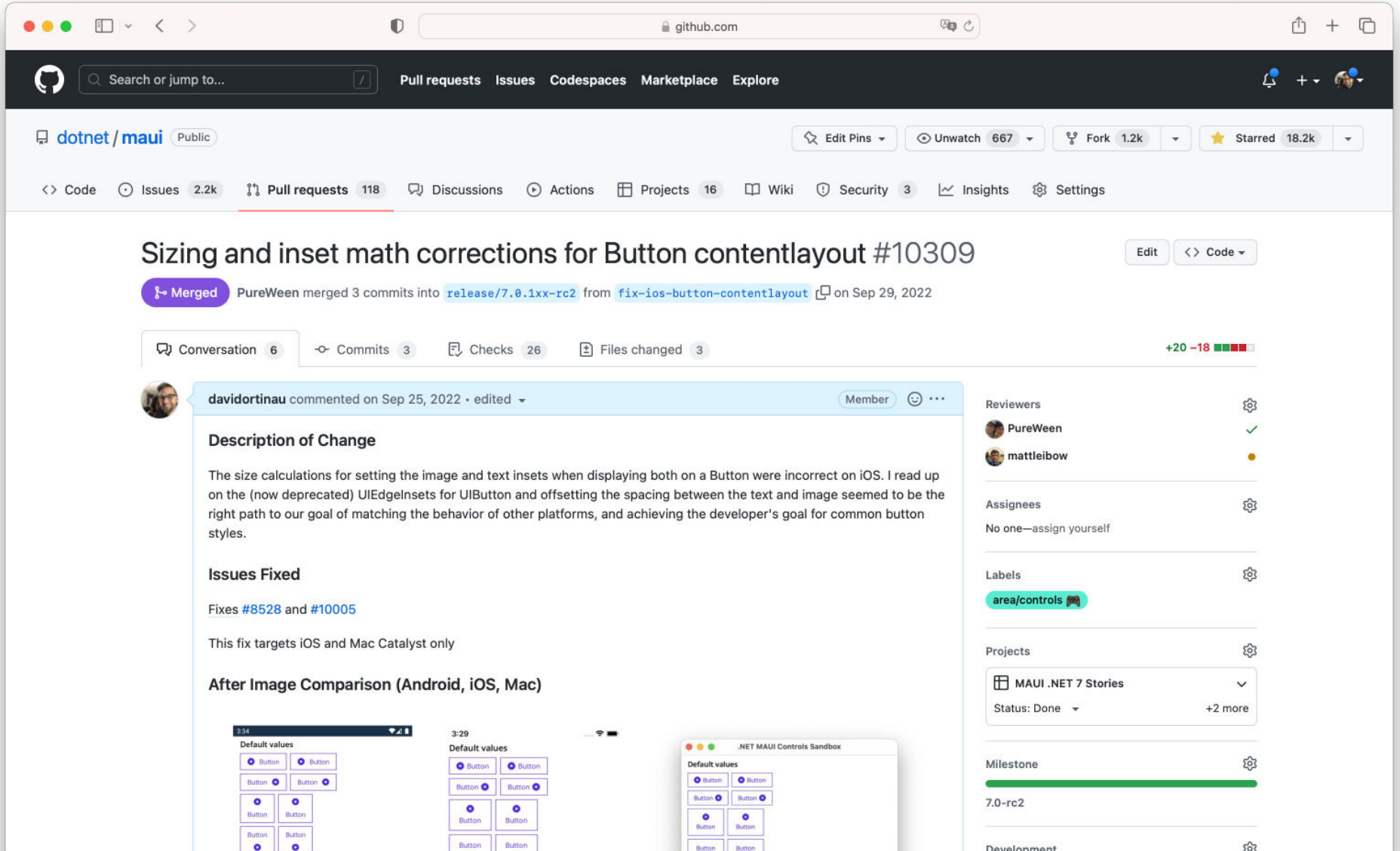
Handlers can be customized to augment the appearance and behavior of a cross-platform control beyond the customization that's possible through the control's API. This customization, which modifies the native views for the cross-platform control, is achieved by modifying the mapper for a handler with one of the following methods:

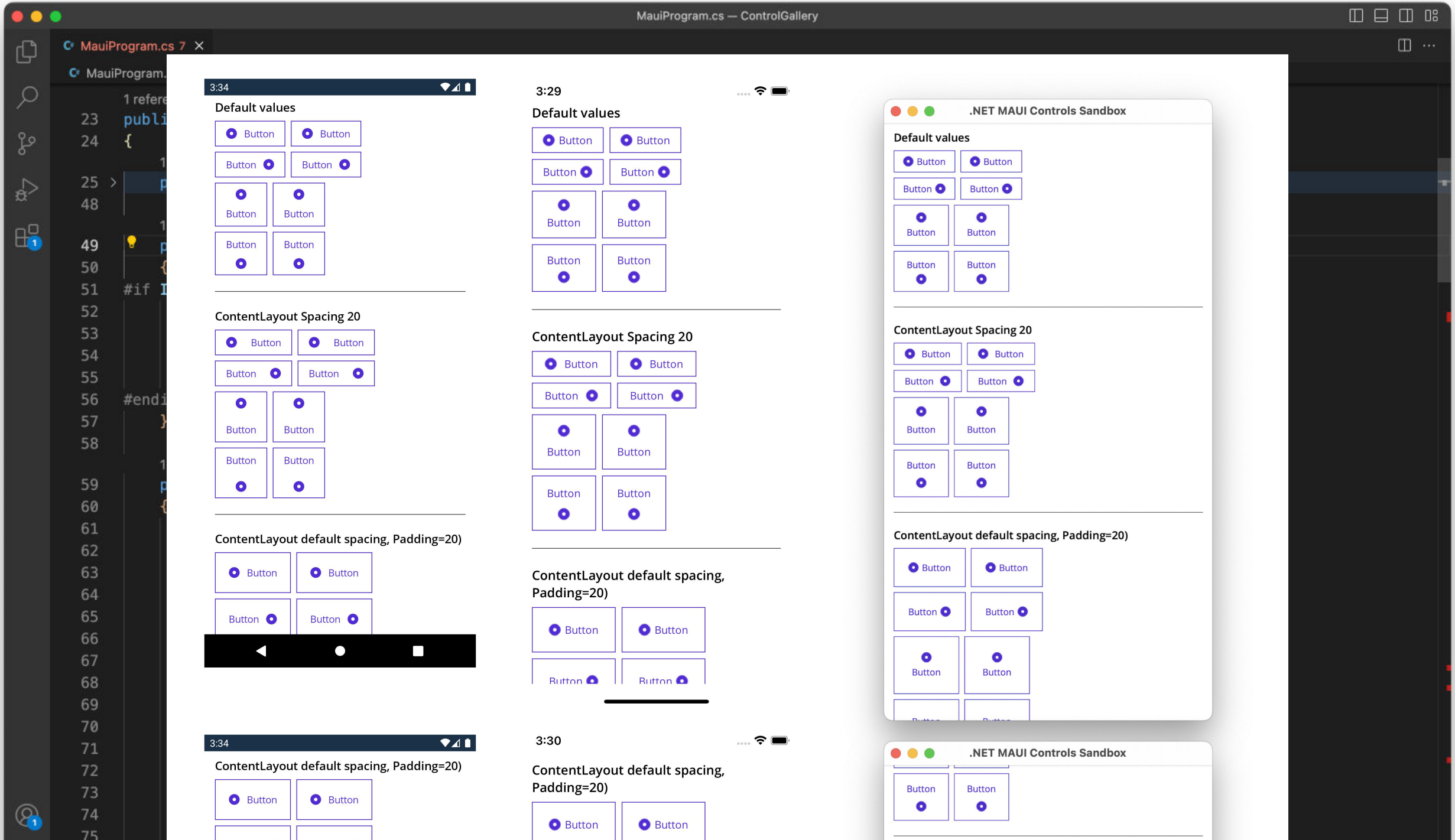
PrependToMapping, which modifies the mapper for a handler before the .NET MAUI control mappings have been applied.

ModifyMapping, which modifies an existing mapping.

AppendToMapping, which modifies the mapper for a handler after the .NET MAUI control mappings have been applied.

A Bug's Story


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Fork 1.2k
Starred 18.2k
Code
Issues 2.2k
Pull requests 118
Discussions
Actions
Projects 16
Wiki
Security 3
Insights
Settings
Sizing and inset math corrections for Button contentlayout #10309
Edit
Code
Merged
PureWeen merged 3 commits into release/7.0.1xx-rc2 from fix-ios-button-contentlayout on Sep 29, 2022
Conversation 6
Commits 3
Checks 26
Files changed 3
+20 −18
davidortinau commented on Sep 25, 2022 • edited
Member
Description of Change
The size calculations for setting the image and text insets when displaying both on a Button were incorrect on iOS. I read up on the (now deprecated) UIEdgeInsets for UIButton and offsetting the spacing between the text and image seemed to be the right path to our goal of matching the behavior of other platforms, and achieving the developer's goal for common button styles.
Issues Fixed
Fixes #8528 and #10005
This fix targets iOS and Mac Catalyst only
After Image Comparison (Android, iOS, Mac)
Default values
Button
.NET MAUI Controls Sandbox
Reviewers
PureWeen
mattleibow
Assignees
No one—assign yourself
Labels
area/controls
Projects
MAUI .NET 7 Stories
Status: Done
+2 more
Milestone
7.0-rc2
Development


MauiProgram.cs — ControlGallery
MauiProgram.cs 7
MauiProgram.
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23 publi
24 {
25 >
48
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51 #if I
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73
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75
3:34
Default values
Button
ContentLayout Spacing 20
ContentLayout default spacing, Padding=20)
3:29
Default values
ContentLayout Spacing 20
ContentLayout default spacing, Padding=20)
.NET MAUI Controls Sandbox
Default values
ContentLayout Spacing 20
ContentLayout default spacing, Padding=20)
3:34
ContentLayout default spacing, Padding=20)
3:30
ContentLayout default spacing, Padding=20)
.NET MAUI Controls Sandbox

Android

Type some text

iOS

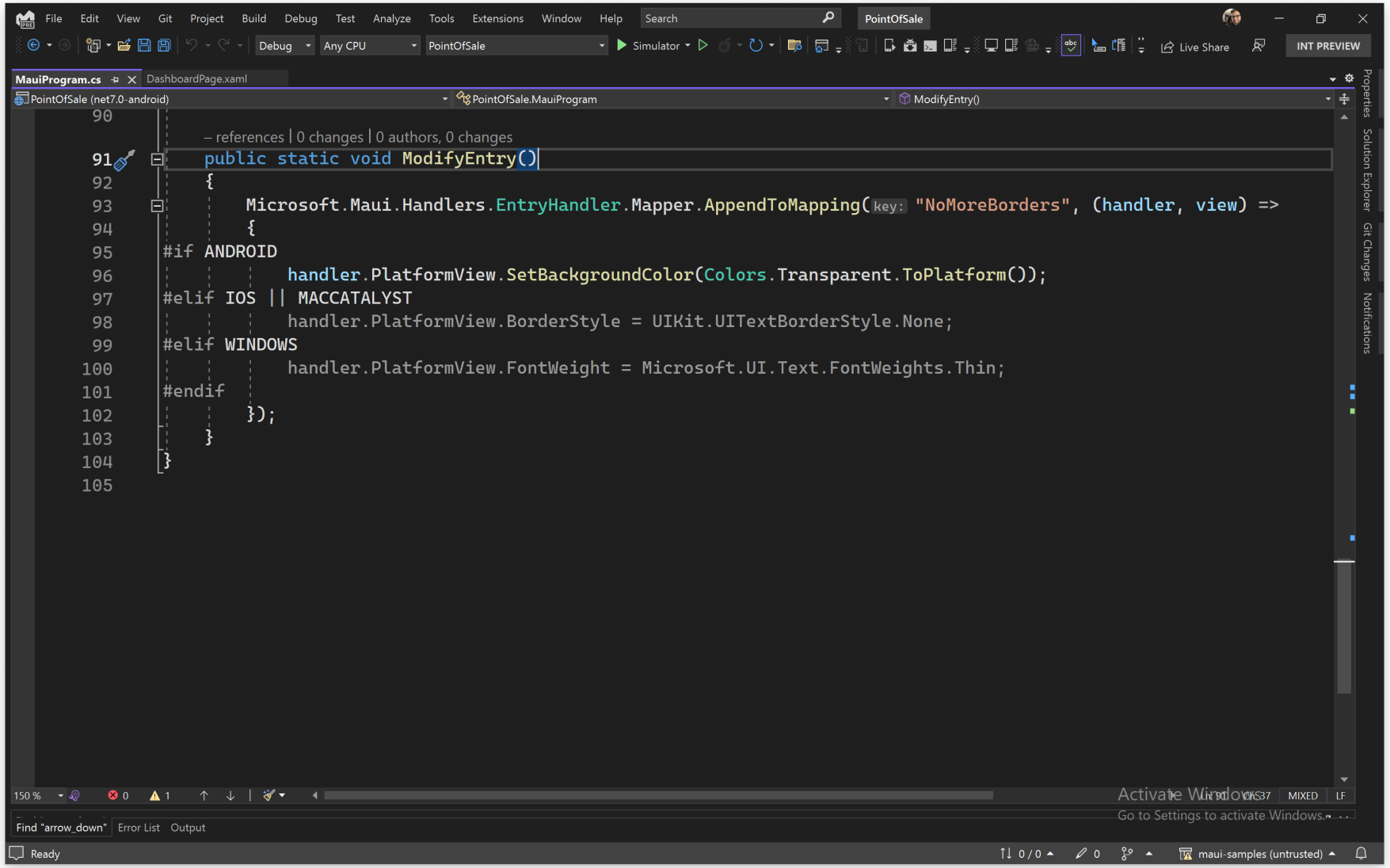
Type some text

macOS

Type some text

Windows

Type some text



Hot Reload

.NET Hot Reload

- Implement MetaDataUpdateHandler to trigger UI refresh
- XAML Hot Reload needs to still be enabled

```
#if DEBUG
[assembly: System.Reflection.Metadata.MetadataUpdateHandler(typeof(VolumeMixer.HotReloadService))]
namespace VolumeMixer
{
    3 references | 0 changes | 0 authors, 0 changes
    public static class HotReloadService
    {
#pragma warning disable CS8632 // The annotation for nullable reference types should only be used in
        public static event Action<Type[]?>? UpdateApplicationEvent;
#pragma warning restore CS8632 // The annotation for nullable reference types should only be used in

        0 references | 0 changes | 0 authors, 0 changes
        internal static void ClearCache(Type[]? types) { }
        0 references | 0 changes | 0 authors, 0 changes
        internal static void UpdateApplication(Type[]? types)
        {
            //if ((types != null && types.Length > 0 && types[0] != null))
            //{
            UpdateApplicationEvent?.Invoke(types);
            //}
        }
    }
}
#endif
```

.NET Hot Reload

- Handle event and trigger rebuild of UI

```
2 references | David Ortinau, 2 days ago | 1 author, 1 change
void Build() ...

0 references | David Ortinau, 2 days ago | 1 author, 1 change
protected override void OnNavigatedTo(NavigatedToEventArgs args)
{
    base.OnNavigatedTo(args);

    BindingContext = new MainViewModel();

    Build();

#if DEBUG
    HotReloadService.UpdateApplicationEvent += ReloadUI;
#endif
}
0 references | David Ortinau, 2 days ago | 1 author, 1 change
protected override void OnNavigatedFrom(NavigatedFromEventArgs args)
{
    base.OnNavigatedFrom(args);

#if DEBUG
    HotReloadService.UpdateApplicationEvent -= ReloadUI;
#endif
}

2 references | David Ortinau, 2 days ago | 1 author, 1 change
private void ReloadUI(Type[] obj)
{
    MainThread.BeginInvokeOnMainThread(() =>
    {
        Build();
    });
}
}
```

Thanks! Let’s talk.

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